Edge and Flow Guided Iterative CNN for Remote Sensing Image Change Detection

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***Abstract*— Change detection (CD) is an essential aspect of urban planning and resource management. Deep learning (DL) has the potential to detect complex changes from massive data more automatically than traditional methods. However, current DL-based CD methods have limited abilities to efficiently extract bi-temporal features, emphasize real changes, detect weak edges, and ultimately integrate multiscale change information in detail. To address these challenges, we propose an edge and flow guided iterative convolutional neural network (EFICNN). Our network introduces an innovative and efficient iterative backbone (IB) as the feature extractor through pretrained fine-tuning. By embedding the IB into a Siamese architecture, it is possible to extract richer bi-temporal features from the input remote sensing (RS) images. On this basis, we design a 3-D difference enhancement module (3D-DEM) that utilizes a parameter-free 3-D attention mechanism to emphasize and connect bi-temporal differential and concatenated change features. Additionally, an edge-guided attention module (EGAM) is developed to enhance weak edges. This module combines reverse attention and edge attention based on the Laplacian pyramid to capture change backgrounds and high-frequency change edges, respectively. Inspired by the optical flow alignment between adjacent frames, we ultimately employ a flow-guided fusion module (FGFM) to promote the propagation of fine-grained features from deep to shallow and dynamically optimize the multiscale fusion process. Qualitative and quantitative experiments on three publicly available datasets demonstrate that our method outperforms 17 SOTA methods in terms of real change recognition and edge refinement. Our code is available at https://github.com/LYT-work/EFICNN.**

***Index Terms*— 3-D difference enhancement module (3D-DEM), change detection (CD), edge-guided attention module (EGAM), flow-guided fusion module (FGFM), iterative backbone (IB).**

I. Introduction

DRIVEN by population growth and rapid urbanization, natural resources and the environment are under immense pressure, leading to pronounced surface changes. In this challenging context, remote sensing (RS) has emerged as a vital tool to provide Earth observation data for detecting changes and promoting sustainable development. RS change detection (CD) involves the comparison of RS images acquired over the same geographical area at different time points. This technique finds various applications in resource management [1], environmental monitoring [2], and urban planning [3]. Advancements in RS technologies have enabled the acquisition of high-resolution spatial and temporal images from diverse sensors. This progress demands CD methods that can effectively handle massive data, deliver superior performance, and provide rapid responses.

Traditional CD methods are generally classified into two primary categories: pixel-based CD (PBCD) and object-based CD (OBCD). PBCD operates directly on pixel values and encompasses methods such as image algebra (e.g., band difference [4], band ratio [5], spectral angle mapper [6]), transformations (e.g., change vector analysis [7], principal component analysis [8], Fourier transform [9]), and postclassification comparison [10]. These methods are characterized by their low computational complexity and ease of implementation. In contrast, OBCD methods [11], [12], [13] first segment bi-temporal images into homogeneous objects based on spectral similarity and spatial adjacency, and then detect changes by comparing these objects. This kind of method is more effective for handling diverse land cover types and varying scales of change. However, both PBCD and OBCD methods rely on manually designed features, which limits their generalizability and robustness.

Deep learning (DL) offers an innovative solution for CD by automatically extracting features from massive data without manual design. Convolutional neural networks (CNNs) are extensively recognized in DL for their effectiveness in image analysis. By stacking multiple convolutional and pooling layers, CNNs can learn hierarchical features while capturing local details, such as edges and textures. CNN-based networks like U-Net [14], UNet++ [15], and DeepLabv3+ [16] have demonstrated high efficacy in CD tasks [17], [18], [19]. In terms of network framework, the encoder–decoder architecture is widely adopted in DL. The encoder first extracts features from the input image using the network's intrinsic feature extraction modules. Pretrained backbone networks such as VGG [20], ResNet [21], and MobileNetV2 [22] are also serve as encoders. These networks, trained on extensive datasets, offer robust feature extraction capabilities that further

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enhance CD performance. The decoder then restores the features extracted by the encoder to the original resolution, ultimately producing CD results.

Currently, the encoder–decoder architectures for CD are broadly classified into single-branch and dual-branch designs. Single-branch architectures typically take the ratio [23], difference [24], or concatenated images [25] as inputs. While simple and easy to implement, single-branch architectures underutilize the information available at each time point. In contrast, dual-branch architectures gather more comprehensive information through parallel encoders. Each encoder can independently extract features from individual temporal images, which are later combined within the network. The Siamese network, a dual-branch model with shared weights and parameters, is extensively used in CD tasks [18], [26], [27], [28], [29].

Most Siamese networks have shown promising performance in detecting bi-temporal changes, but they still encounter several challenges. During bi-temporal feature extraction, pretrained backbone networks generally converge more quickly and achieve better performance through fine-tuning, compared to networks trained from scratch [26], [28], [29]. Deeper and broader backbones capture more detailed multilevel features but come with a higher parameter count. Designing a new backbone requires substantial expertise and computational resources for pretraining on large datasets. Therefore, selecting an appropriate backbone for CD involves a trade-off between performance and efficiency. Following feature extraction, the generation of difference feature maps also impacts CD results. Common methods include feature difference [24], [30], [31] and feature concatenation [17], [18], [25]. The former emphasizes change contrast but is sensitive to imaging and registration conditions, while the latter contains more temporal information but does not directly highlight changes. To address the challenges illustrated in Fig. 1, some studies [26], [27], [32] have incorporated attention mechanisms such as channel attention module (CAM) [33], convolutional block attention module (CBAM) [33], and self-attention [34] in these branches to highlight real changes and resist pseudo changes. Nevertheless, these methods tend to increase computational complexity and capture channel and spatial information separately. Furthermore, edges are crucial for detecting overall real changes. Detecting weak edges remains challenging in scenes with dense building distribution and shadows. While multiscale feature fusion methods [35], [36], [37] can refine edges, they are susceptible to noise and background interference. Attention-based edge detection methods [28], [38], [39] can suppress such interference, but they often lack interpretability and clear physical meaning. Before outputting final CD results, most CD networks employ bottom-up sample strategies [24], [29], [40] to fuse deep and shallow change features. However, this approach fails to account for semantic differences and spatial alignment between features of different resolutions after a series of optimizations, resulting in accumulated change information loss during feature fusion process [41].

Given the challenges outlined above, we propose an end-to-end Siamese network for CD, namely edge and flow guided iterative CNN edge and flow guided iterative CNN (EFICNN).

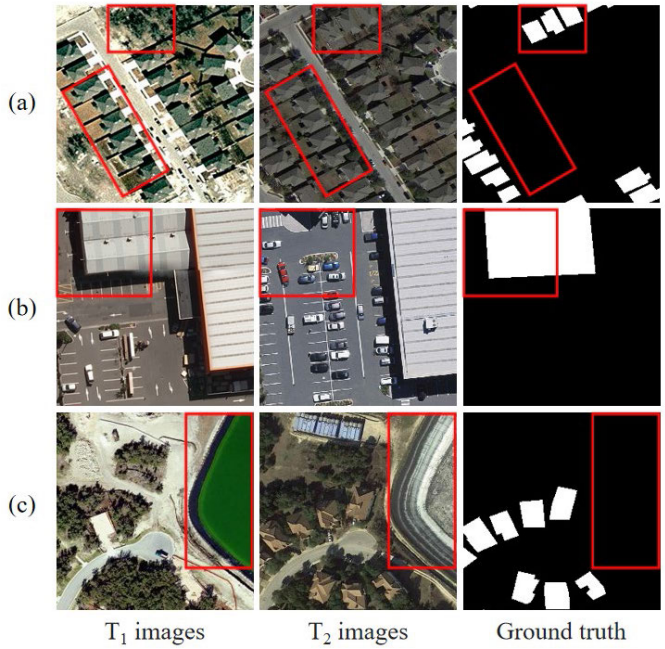


Fig. 1. Various challenges in RS image CD. (a) Spectral inconsistency and weak changes. (b) Changes under shadows. (c) Pseudo changes.

This network aims to improve CD performance by enhancing feature expression and edge detection. Our contributions can be summarized as follows.

1) We introduce a novel and efficient iterative backbone (IB), constructed by two parallel identical VGG16 with a pyramid structure. This design integrates multiscale composite features from two backbones without additional pretraining. When embedded in a Siamese architecture, it significantly captures richer bi-temporal features compared to a single backbone.

2) We design a 3-D difference enhancement module (3D-DEM) that improves the connection between difference and concatenated branches by interactively computing spatial and channel information through a 3-D attention. Without attention parameters, this module facilitates the detection of real changes and resists pseudo changes.

3) We explore an edge-guided attention module (EGAM) that utilizes reverse attention to detect background changes and Laplacian pyramid attention to emphasize high-frequency edges while reducing noise. Applying this module to multiscale features effectively enhances weak edges and improves change classification.

4) We develop a flow-guided fusion module (FGFM) to integrate multiscale features. It aligns feature maps at both adjacent-level and cross-level using optical flow estimation, promoting fine-grained feature propagation from deep to shallow levels. Additionally, a gating mechanism (GM) in this module is introduced to further dynamically optimize the feature fusion process.

The rest of this article is structured as follows. Section II reviews related work. Section III elaborates on the proposed network architecture. Experimental methodology and results

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are presented in Section IV. Section V contains a discussion of the findings. Finally, Section VI concludes our study.

II. Related Works

A. DL-Based CD Methods

CNNs offer unique advantages in image analysis, making them well-suited for RS Daudt et al. [17] introduced three fundamental models based on fully convolutional networks (FCNs) [42] that identify changes through simple feature concatenation or difference. To mitigate error accumulation from feature differences, Peng et al. [25] presented an end-to-end network utilizing an improved UNet++. Methods incorporating attention mechanisms and multiscale feature fusion have further improved CD performance. Fang et al. [18] introduced a densely connected Siamese network, built on UNet++ and enhanced with CAM to preserve fine-grained features. Zhang et al. [26] proposed a deeply supervised image fusion network that utilized spatial and channel attention to improve feature representations. Han et al. [32] proposed a lightweight and efficient self-attention mechanism to integrate and refine multiscale features. In addition, several different enhancement strategies have been employed to optimize the CD networks. Zhang et al. [24] devised a differential pyramid for extracting multiscale difference feature maps. Lei et al. [27] developed a DEM that utilizes CAM and residual connections to suppress irrelevant noise and emphasize real changes. Chen and Shi [30] generated difference maps by computing the Euclidean distance between features, effectively suppressing pseudo changes. Moreover, lightweight networks are applied in CD to optimize the balance between performance and efficiency. Lei et al. [31] optimized feature extraction efficiency and accuracy using multiscale decoupled convolution and spatial-spectral feature cooperation strategies. Li et al. [29] enhanced feature aggregation and change identification using mobile networks, achieving excellent results with low computational costs.

Advancements in transformer-based CD networks have shown their capability in modeling long-term dependencies. Chen et al. [43] pioneered a method that converts bi-temporal images into semantic tokens and utilizes a transformer framework to model their relationships. Bandara and Patel [44] proposed a transformer-based CD network featuring a Siamese transformer encoder and a multilayer perceptron (MLP) decoder. Yan et al. [45] introduced a fully transformer network for CD, which improves feature representation through global perspective extraction and pyramid fusion. Zhang et al. [46] designed a fully transformer network with a Siamese U-shaped architecture to overcome CNNs limitation in capturing global changes. Additionally, hybrid methods combining CNNs and transformers are gaining increasing prominence. Xia et al. [38] proposed an edge-guided network that integrates ResNet and transformer encoders. Li et al. [47] combined transformers with U-Net to extract both global and detailed change information. Furthermore, large foundation models such as vision transformer (ViT) [48] and CLIP [49] have achieved remarkable success in computer vision, and also demonstrated strong generalization in CD [50], [51]. Not long ago, Chen et al. [52] advanced the application of the Mamba [53] across three distinct CD tasks, outperforming CNNs and transformers by integrating spatiotemporal modeling.

Transformers excel at modeling long-range dependencies, but face significant computational complexity as the number of tokens increases. Mambas offer the advantage of high-speed computation but suffer from issues such as gradient vanishing or explosion in large networks. Additionally, with similar model efficiency, Mamba sometimes underperforms relative to advanced CNNs, transformers, and hybrid models [54]. Given its superior stability and ability to capture spatial details, we adopt CNN as the foundational network and optimize feature extraction based on this architecture.

B. Edge-Guided CD Networks

Edges reflect contour and texture features in RS images, playing a crucial role in CD. Traditional edge detection methods, such as Canny and Sobel operators, rely on gradients of local pixel grayscale values. In contrast, DL-based methods can learn more abstract and complex edge features. Current edge-guided CD networks can be categorized into two categories: feature fusion [35], [36], [37] and attention-guided methods [28], [38], [39]. The former extracts edge and nonedge features separately before integrating them. The latter employs attention mechanisms to recognize the importance of edges and combine them with other features. Despite their potential, edge cues remain underutilized in CD tasks. Bai et al. [37] integrated discriminative information and edge structure before feeding it into a Siamese CNN to generate more accurate building boundaries and change maps. Chen et al. [36] utilized a hierarchical transformer combined with an edge-aware module to enhance both internal changes and edge details. Zhang et al. [28] employed an edge refinement module to perceive and refine the change edges. Zhu et al. [39] introduced an edge-aware module and an edge-guided feature module with CAM to enhance blurred edge representation. To simultaneously capture both nonedge and edge changes, we propose the EGAM and apply it on multiscale features. This module is built on Laplacian pyramid [55] edge enhancement operator and integrates it into the network's forward propagation, providing an intuitive physical interpretation.

C. Optical Flow-Guided CD Networks

Optical flow, derived from physical optical phenomena, estimates displacement vectors for each pixel between consecutive frames. Traditional optical flow estimation methods such as Lucas–Kanade [56], Horn–Schunck [57], and correlation-based approaches [58], rely on the brightness constancy and spatial smoothness assumptions. In contrast, DL-based methods like FlowNet [59], PWC-Net [60], and SpyNet [61] have significantly advanced the field. Optical flow estimates pixel motion between consecutive frames, similar to how semantic flow captures information across different scales. Currently, optical flow is primarily used in CD for precise image registration when constructing temporal differences. For instance, Fang et al. [62] introduced a dual-alignment fusion module to correct misalignments in multitemporal images, and combined feature interaction and fusion strategies to generate difference maps. Zhang et al. [63] employed the alignment module based

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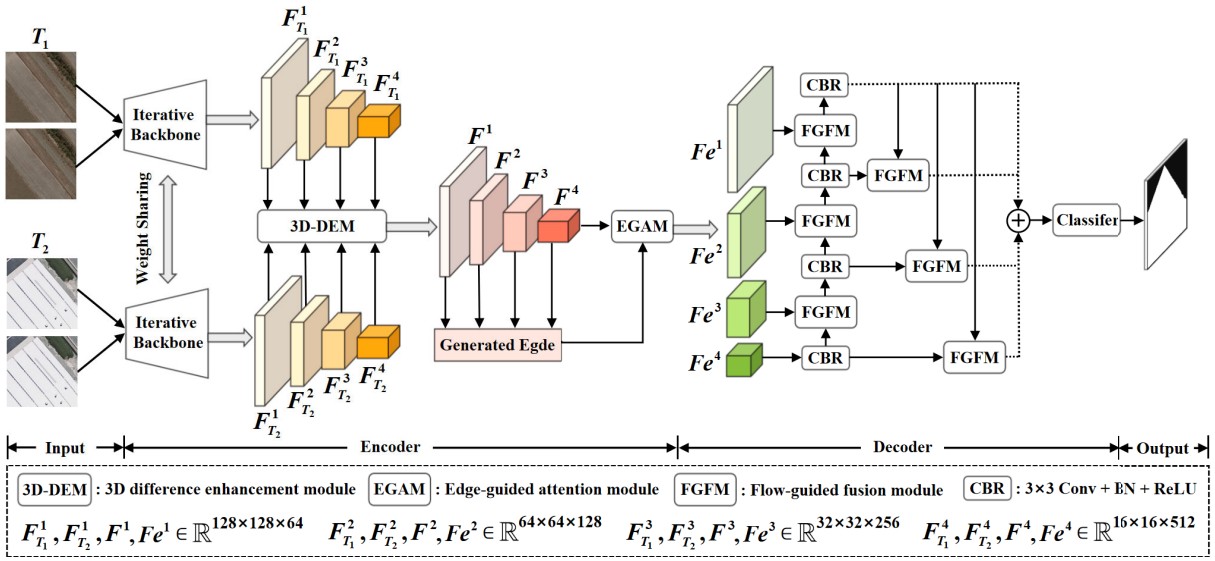


Fig. 2. Overall architecture of our proposed network. The yellow blocks represent multiscale features extracted by the IB, the pink blocks denote the difference-enhanced features, and the green blocks signify the edge-enhanced difference maps. The intensity of the color increases with the depth of the feature.

on differential flow field to estimate spatial offsets between bi-temporal images, ensuring spatial alignment and addressing registration deficiencies from perspective differences. Li et al. [64] utilized optical flow information to resample bi-temporal features, correcting vertical displacements and misalignments in different directions for building Tian et al. [65] designed a deformable convolution unit to learn offsets for aligning bi-temporal deep features before generating accurate difference maps. In addition to bi-temporal registration, Feng et al. [66] proposed a novel approach for change feature fusion using optical flow. This method generated displacement maps through optical flow estimation and aligned adjacent-level features using learnable affine transformations. During the progressive fusion process, it effectively improved the model's spatial invariance and robustness. Unlike previous studies, our study focuses on the alignment and integration of adjacent- and cross-level features to address semantic misalignment and invalid propagation between deep and shallow features after optimization, thus improving the multiscale fusion process.

III. Methodology

This section begins with an introduction to the overall architecture of our EFICNN. Following this, detailed descriptions of each module in the network are provided.

A. Overview

The proposed CD network is a weight-sharing encoder-decoder design, depicted in Fig. 2. The network consists of four key innovative components: IB, 3D-DEM, EGAM, and FGFM. In the encoder, each IB includes two identical VGG16 with a pyramid structure that simultaneously receives the same inputs at a single time point and extracts multiscale composite features through iterative fusion. When embedded into a Siamese structure, the IB captures richer bi-temporal features at four different levels compared to a single backbone. Subsequently, the feature maps generated at each level are fed into the 3D-DEM to further produce difference maps. The 3D-DEM consists of two branches: a difference branch and a concatenation branch. To capture interactive channel and spatial changes at low cost, these two branches are combined with a parameter-free 3-D attention to highlight real changes and resist pseudo changes. The enhanced difference feature map at each level and its generated edge are then fed into EGAM. This module utilizes reverse attention to identify background changes and Laplacian pyramid attention to emphasize high-frequency change edges. In the decoder, FGFM is employed to align and fuse feature maps at different resolutions, enabling both adjacent-level and cross-level integration of deep and shallow features. Unlike interpolation-based methods that rely on uniform pixel neighborhoods, FGFM leverages displacement information from optical flow estimation to achieve feature alignment, and dynamically adjusts the fusion ratio through a GM. Finally, the classifier generates single-channel binary CD results through size recovery and channel compression.

B. Iterative Backbone

Many studies have shown that using a pretrained CNN backbone can enhance model convergence and improve CD performance [28], [29], [30]. Given the computational costs associated with designing new backbones and extracting deeper and larger features, we propose an iterative VGG16 [20] pretrained on ImageNet [67] to optimize bi-temporal feature extraction.

VGG16 is a classic architecture in CNNs, constructed by stacking convolutional layers, pooling layers, and fully connected layers to achieve deep feature extraction and classification. Its simplicity and modularity facilitate easy fine-tuning. The convolutional component of VGG16 consists of five convolutional blocks, each containing multiple 3×3 convolutional layers, followed by ReLU activation functions.

Additionally, each convolutional block is followed by a max pooling layer, which reduces the spatial resolution of the

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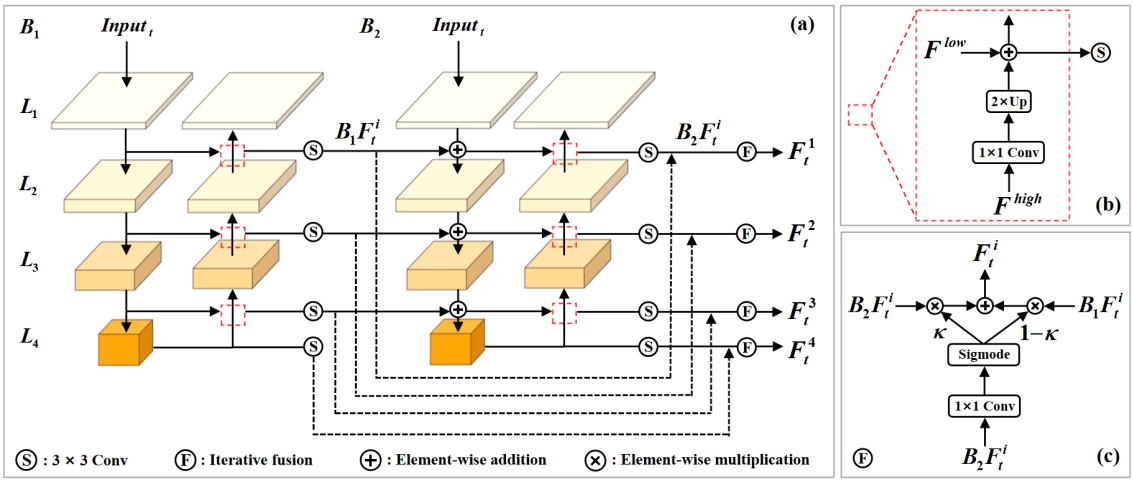


Fig. 3. Iterative VGG16 backbone. (a) Iterative structure. (b) Feature enhancement. (c) Iterative fusion.

TABLE I
SUB-VGG16 STRUCTURE

Level	Input Size	Operation	Output Size
L_1	256 × 256 × 3	[3 × 3, 64 Conv + ReLU; 3 × 3, 64 Conv + ReLU] × 2 Max Pooling	128 × 128 × 64
L_2	128 × 128 × 64	[3 × 3, 128 Conv + ReLU; 3 × 3, 128 Conv + ReLU] × 2 Max Pooling	64 × 64 × 128
L_3	64 × 64 × 128	[3 × 3, 256 Conv + ReLU; 3 × 3, 256 Conv + ReLU] × 3 Max Pooling	32 × 32 × 256
L_4	32 × 32 × 256	[3 × 3, 512 Conv + ReLU; 3 × 3, 512 Conv + ReLU] × 3 Max Pooling	16 × 16 × 512

feature maps. Considering the computational cost, we treat each convolutional block along with its corresponding max pooling layer as a basic block and utilize the feature maps output from the first four such blocks. As detailed in Table I, the network extracts multiscale features at four levels: L_1, L_2, L_3, and L_4. Let f_t^i ($t = T_1, T_2$ and $i = 1, 2, 3$) represents the output feature of the ith level from a single VGG16 backbone at time t. Accordingly, f_t^{i+1} can be expressed as

$$f_t^{i+1} = \text{BasicBlock}\left(f_t^i\right) \tag{1}$$

where BasicBlock includes the convolutional block and max pooling operation.

Differently, the IB shown in Fig. 3(a) consists of two identical subnetworks, denoted as B_1 and B_2, both initialized with weights from an open-source pretrained VGG16. In B_1, four feature maps are first generated through the aforementioned VGG16 single backbone. Subsequently, an FPN [68]-like structure is applied to enhance the multiscale representation of features. As illustrated in Fig. 3(b), for high-level features in B_1, a 1 × 1 convolution is first used to adjust the channel dimensions. The upsampled high-level feature is then added to the adjacent lower-level feature, progressively propagating upward. To refine and smooth the accumulated features, a 3 × 3 convolution is applied at each level, ultimately producing the output feature maps of B_1, denoted as $B_1F_t^i$ ($t = T_1, T_2$ and $i = 1, 2, 3$). This process can be expressed as

$$\begin{aligned} B_1F_t^{i+1} &= \text{BasicBlock}\left(B_1F_t^i\right) \\ B_1F_t^i &= \text{Conv}_{3\times3}\left(\text{Up}\left(\text{Conv}_{1\times1}\left(B_1F_t^{i+1}\right)\right) + B_1F_t^i\right) \end{aligned} \tag{2}$$

where $\text{Conv}_{3\times3}$ and $\text{Conv}_{1\times1}$ represent convolutional layers with kernel sizes of 3 × 3 and 1 × 1, respectively. Up(·) refers to feature upsampling by a factor of 2.

In B_2, the input at each level includes not only the output from the previous level but also the feature from the parallel level in B_1. These features are summed and propagated downward through the same VGG16 network. Similar to B_1, after multiscale feature enhancement, the final output of B_2 is $B_2F_t^i$ ($t = T_1, T_2$ and $i = 1, 2, 3$), expressed as

$$\begin{aligned} B_2F_t^{i+1} &= \text{BasicBlock}\left(B_1F_t^i + \text{B}_2F_t^i\right) \\ B_2F_t^i &= \text{Conv}_{3\times3}\left(\text{Up}\left(\text{Conv}_{1\times1}\left(B_2F_t^{i+1}\right)\right) + B_2F_t^i\right). \end{aligned} \tag{3}$$

Note that B_1 and B_2 share weights to prevent unnecessary parameter increases and ensure consistency in the feature space. Despite sharing weights and having the same structures, B_2 is capable of capturing composite features. Moreover, $B_1F_t^4$ and $B_2F_t^4$ do not involve multiscale feature enhancement and are instead obtained through a 3 × 3 convolution.

To further facilitate iterative fusion, a self-attention mechanism is introduced, as depicted in Fig. 3(c). This mechanism dynamically computes the weight κ for $B_2F_t^i$, while the remaining weight $1-\kappa$ is assigned to $B_1F_t^i$, thereby emphasizing the importance of $B_2F_t^i$. By performing a weighted summation of these two feature maps, the final output F_t^i ($t = T_1, T_2$ and $i = 1, 2, 3, 4$) of the IB is obtained. The iterative fusion can be expressed as

$$\begin{aligned} \kappa &= \sigma\left(\text{Conv}_{1\times1}\left(B_2F_t^i\right)\right) \\ F_t^i &= \kappa \times B_2F_t^i + (1-\kappa) \times B_1F_t^i \end{aligned} \tag{4}$$

where $\sigma(\cdot)$ indicates the Sigmoid function.

In our CD network, a Siamese dual-branch architecture is employed to separately process inputs from earlier and later time points. Each branch utilizes the IB to extract composite

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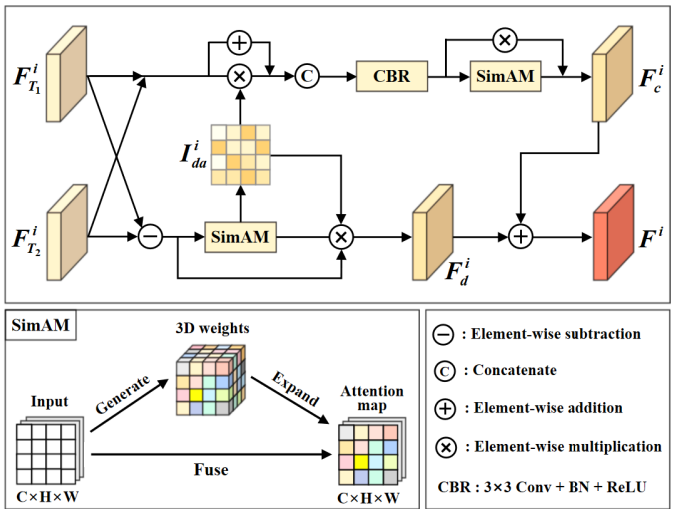


Fig. 4. Illustration of the 3D-DEM. In this module, the SimAM attention mechanism generates 3-D weights from the input feature. These weights are then expanded into a 3-D attention map, which is fused with the original input to highlight the salient regions.

features for each time point. Given two inputs with dimensions $\mathbb{R}^{H\times W\times C}(H = 256,\ W = 256,\ C = 3)$, where H and W, respectively, denote the height and width, C denotes the number of channels. As shown in Fig. 2, after processing through the IB, the composite output $F^i_{T_1}$ and $F^i_{T_2}$ are generated. These multiscale features from T_1 and T_2 are subsequently compared to produce the difference feature maps.

C. 3-D Difference Enhancement Module

To further capture changes, most networks employ feature difference, feature concatenation, or fusion methods [40], [62], [69]. Particularly, fusion methods combine the advantages of feature difference and concatenation, highlighting change contrast while preserving bi-temporal information. Building on this, the introduction of attention mechanisms can further emphasize changes and suppress noise, though this may increase computational complexity. In addition, many existing attention mechanisms separately operate in spatial and channel dimensions, failing to consider their interactions. Therefore, we propose the 3D-DEM, as shown in Fig. 4. This module integrates a parameter-free 3-D attention mechanism to compute 3-D weights for each neuron, facilitating a more effective connection between the two branches.

Before delving into the two feature branches, it is essential to provide an overview of the 3-D attention mechanism, SimAM [70]. Unlike traditional channel and spatial attention mechanisms, SimAM is a parameter-free attention mechanism that assigns 3-D attention weights to each element (i.e., each neuron) in the feature map. These weights account for the importance of neurons across three dimensions: channel, height, and width. Theoretically, SimAM is founded on a neuroscientific hypothesis that activated neurons generally suppress their surrounding neurons. The most straightforward method for identifying activated neurons is to assess the linear separability from their surroundings. To this end, SimAM employs a minimal energy function to measure this linear separability, defined as $e_\alpha(x)$

$$\begin{aligned}\hat{\mu} &= \frac{1}{H\times W}\sum_{i=1}^{H\times W}\beta\\ \hat{\sigma}^2 &= \frac{1}{H\times W}\sum\nolimits_{i=1}^{H\times W}\left(\beta-|\hat{\mu}\right)^2\\ e_\alpha(x) &= \frac{4\left(\hat{\sigma}^2+\varphi\right)}{\left(\alpha-|\hat{\mu}\right)^2+2\hat{\sigma}^2+2\varphi}.\end{aligned}\tag{5}$$

Given an input tensor $x \in \mathbb{R}^{H\times W\times C}$, the variables α and β refer to the target neuron and surrounding neurons in a single channel of x. The symbols $\hat{\mu}$ and $\hat{\sigma}^2$ represent the mean and variance calculated over all neurons of x along the channel dimension. To ensure numerical stability, a small constant φ with a value of $1e^{-4}$ is included. According to (5), lower energy values indicate greater distinction between the target neuron and its neighbors, signifying stronger spatial suppression and higher importance. Consequently, the importance of neurons can be quantified as $1/e_\alpha(x)$. After determining the importance, the Sigmoid function $\sigma(\cdot)$ is applied to constrain excessively large values, ultimately yielding the 3-D weights as $\sigma(1/e_\alpha(x))$.

In the feature difference branch, the elementary difference feature map F^i_{ed} is generated by element-wise subtraction between $F^i_{T_1}$ and $F^i_{T_2}(i=1,2,3,4)$. To further highlight the channel and spatial differences, the SimAM attention mechanism is employed to compute 3-D attention weights for the elements within F^i_{ed}. This computation yields a difference attention map I^i_{da}, which quantifies the degree of change. Subsequently, I^i_{da} is element-wise multiplied with F^i_{ed} to derive the enhanced difference map F^i_d, mathematically expressed as

$$\begin{aligned}F^i_{ed} &= \left|F^i_{T_1}-F^i_{T_2}\right|\\ I^i_{da} &= \sigma\left(1/e_\alpha\left(F^i_{ed}\right)\right)\\ F^i_d &= I^i_{da}\times F^i_{ed}\end{aligned}\tag{6}$$

where $F^i_{T_1}$ and $F^i_{T_2}$ represent the bi-temporal features extracted from the IB at the same level.

In the feature concatenation branch, the computed difference attention I^i_{da} is used to refine $F^i_{T_1}$ and $F^i_{T_2}$ to get the difference focus. The refined features are then concatenated along the channel dimension and integrated into a joint representation F^i_j through a CBR convolution block. Similarly, SimAM is used to evaluate the importance of the joint feature and output the enhanced concatenated map F^i_c. To amplify the differences between input features, the outputs of the two branches are added together, yielding the final discriminative difference feature map $F^i(i=1,2,3,4)$. This process can be expressed as

$$\begin{aligned}F^i_j &= \mathrm{CBR}\left(\mathrm{Cat}\left(I^i_{da}\times F^i_{T_1}+F^i_{T_1}, I^i_{da}\times F^i_{T_2}+F^i_{T_2}\right)\right)\\ F^i_{\mathrm{c}} &= \sigma\left(1/e_\alpha\left(F^i_j\right)\right)\times F^i_j\\ F^i &= F^i_d+F^i_{\mathrm{c}}\end{aligned}\tag{7}$$

where CBR is composed of a 3×3 convolutional layer, batch normalization (BN) layer, and ReLU function. $\mathrm{Cat}(\cdot)$ represents feature channel concatenation.

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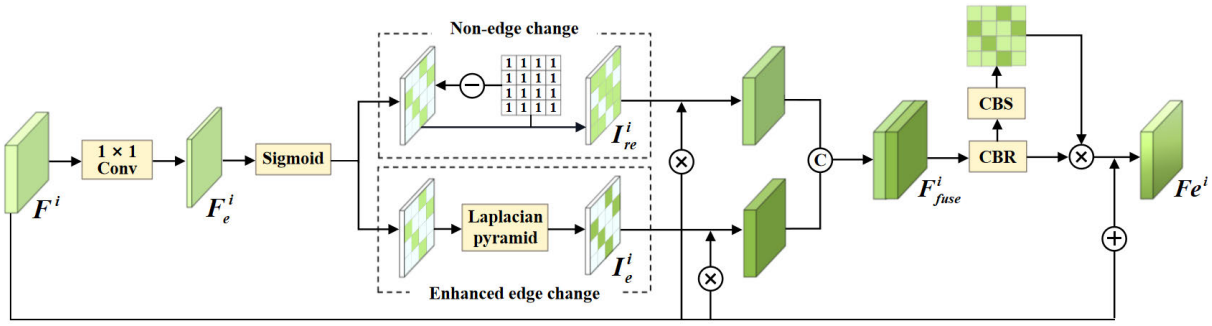


Fig. 5. Illustration of the EGAM.

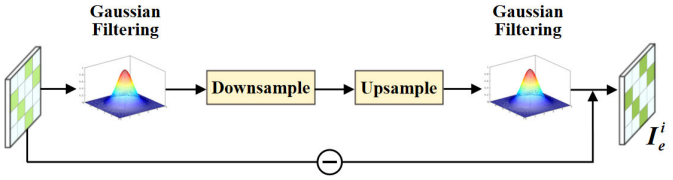


Fig. 6. Illustration of the Laplacian pyramid-based edge attention in EGAM.

D. Edge-Guided Attention Module

Edges indicate regions of rapid intensity variation, providing valuable cues for CD. Due to lighting and background effects, edges in RS images often appear blurred or attenuated. To address this challenge, we introduce the EGAM, which utilizes both change backgrounds and enhanced edges to guide change classification.

The EGAM is based on a combination of reverse attention I^i_{re} and edge attention I^i_e, as shown in Fig. 5. Specifically, the edge attention focuses on the overall contour of the change target, making it susceptible to background interference. In contrast, the reverse attention helps mitigate this interference, directing the network toward the remaining change regions. By combining these two attentions, EGAM provides a comprehensive understanding of change targets in complex scenes. First, the reverse attention I^i_{re} is defined to represent the weights of nonedge changes

$$I^i_{\text{re}} = 1 - \sigma\left(F^i_e\right) \tag{8}$$

where $F^i_e (i = 1, 2, 3, 4)$ denotes the single-channel edge feature generated from the enhanced difference map F^i.

On the other hand, the edge attention is computed based on the Laplacian pyramid. This operator extracts image features at different scales by constructing a Gaussian pyramid and computes the Laplacian pyramid by differencing adjacent layers. It can effectively capture edge information, emphasize high-frequency components, and reduce artifacts. To preserve most high-frequency details while minimizing computational cost, we select the first layer of the Laplacian pyramid and incorporate it into the forward propagation of the network. As shown in Fig. 6, the construction of this layer first requires smoothing the input $\sigma(F^i_e)$ through Gaussian filtering to suppress its high-frequency details. The smoothing operation employs a 2-D 5×5 Gaussian convolution kernel, which assigns the highest weight to the central point and progressively weaker weights to the surrounding pixels. This effectively reduces high-frequency noise in the image while preserving its main features. Next, the smoothed image undergoes downsampling and upsampling operations. Downsampling reduces the image's resolution and computational load by halving the image dimensions in each direction. Upsampling then restores the image to its original resolution by inserting zero values, often combined with Gaussian convolution to blur the image details. Finally, the input and the upsampled image I^i_{up} are pixel-wise subtracted to obtain the difference image I^i_e. This difference image retains the high-frequency information from the original image that is removed during Gaussian filtering and upsampling, effectively highlighting the weights of edge details

$$\begin{aligned} I^i_{\text{up}} &= \text{Ga}\big(\text{Up}\big(\text{Dn}\big(\text{Ga}\big(\sigma\left(F^i_e\right)\big)\big)\big)\big) \\ I^i_e &= \sigma\left(F^i_e\right) - I^i_{\text{up}} \end{aligned} \tag{9}$$

where $\text{Ga}(\cdot)$ denotes the Gaussian convolution operation. $\text{Up}(\cdot)$ and $\text{Dn}(\cdot)$ represent the upsampling and downsampling processes, respectively.

After independently obtaining the reverse attention and edge attention weights shown in Fig. 5, we multiply them separately with the corresponding difference feature map F^i to get the background map and enhanced edge map associated with the changes. The CBR block fuses these two to achieve a more comprehensive enhancement effect. To further reduce the potential noise in F^i_{fuse} that may hinder change classification and enhance feature representation, we introduce an attention mask to focus on critical regions. The final output of EGAM, denoted as Fe^i, is obtained by performing element-wise multiplication of the attention mask with F^i_{fuse}, followed by an addition with F^i

$$\begin{aligned} F^i_{\text{fuse}} &= \text{CBR}\big(\text{Cat}\big(F^i \times I^i_{\text{re}}, F^i \times I^i_e\big)\big) \\ Fe^i &= \text{CBS}\big(F^i_{\text{fuse}}\big) \times F^i_{\text{fuse}} + F^i \end{aligned} \tag{10}$$

where CBR is composed of a 1×1 convolutional layer, BN, and ReLU function. CBS is a set of 1×1 convolutional layer, BN, and Sigmode function. $\text{CBS}(F^i_{\text{fuse}})$ denotes the attention mask of F^i_{fuse}.

E. Flow-Guided Fusion Module

To integrate enhanced features across different scales, it is essential to align low-resolution feature maps with high-resolution ones. Traditional fusion methods typically employ bilinear interpolation to achieve this alignment. However, this

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method relies on uniform pixel neighborhoods, which inadequately account for the semantic discrepancies and spatial relationships between features of different resolutions after optimization by preceding modules. Consequently, it can result in alignment inaccuracies and accumulated change information loss when transferring from deep to shallow levels. To address these challenges and optimize the fusion process, we introduce the FGFM, as shown in Fig. 7. This module comprises three stages: optical flow field generation, optical flow warping, and gated fusion.

In the optical flow field generation stage, inspired by FlowNet-S [59], we establish precise correspondences between feature maps of different resolutions. The process begins with a 1×1 convolution applied to the low-resolution feature map to match its channel dimension with that of the high-resolution feature map. Afterward, a preliminary spatial alignment is performed using bilinear interpolation, followed by channel fusion to integrate the features across two different scales. The optical flow field is then generated through a 3×3 convolution, resulting in a four-channel flow field. The first two channels represent the horizontal and vertical displacements of the high-resolution feature map, while the latter two represent the displacements of the low-resolution feature map. These displacement vectors are crucial for adjusting the spatial positions of the feature maps, ensuring more precise alignment. The process is expressed as follows:

$$Fl = \mathrm{Conv}_{\mathrm{flow}}\left(\mathrm{Cat}\left(Fe_{\mathrm{high}}, \mathrm{Up}(\mathrm{Conv}_{1\times1}(Fe_{\mathrm{low}}))\right)\right) \tag{11}$$

where Fl denotes the optical flow map, Fe_{low} represents the low-resolution deep feature map, and Fe_{high} represents the high-resolution shallow feature map. $\mathrm{Conv}_{\mathrm{flow}}$ and $\mathrm{Conv}_{1\times1}$ refer to the 3×3 and 1×1 convolutional operations, respectively.

In the optical flow warping stage, a grid sampling technique is employed to remap each pixel location of the feature maps according to the displacement information from the optical flow field. Specifically, let h and w denote the height and width of the grid, respectively. Values are uniformly distributed in the range of $[-1, 1]$ along both the vertical and horizontal axes. The Cartesian product of these axes forms the grid Gr, which consists of a set of points $\{(x,y)\}$, where $x = 1, 2, \ldots, h$ and $y = 1, 2, \ldots, w$. To transform the grid based on the optical flow field, we perform the following operations on each point (x,y) in the grid

$$\left(x', y'\right) = (x + u(x, y), y + v(x, y)) \tag{12}$$

where (x,y) denotes the initial coordinates, while (x',y') signifies the transformed coordinates determined by the horizontal component $u(x,y)$ and vertical component $v(x,y)$ of the optical flow field. Once the warped grid $Gr' = \{(x',y')\}$ is obtained, aligned output is generated by sampling the input feature map at these coordinates. Applying optical flow warping individually to high- and low-resolution feature maps facilitates more precise alignment.

In the gated fusion stage, a GM dynamically adjusts the fusion weights of different feature maps, enabling adaptive capture of both global and detailed change information. First, the warped feature maps Fw_{low} and Fw_{high} are added together to obtain the joint feature Fw. Based on Fw, the CAM and Sigmoid function are applied to generate the gating coefficient ω_{gate}. This coefficient controls the weights of Fw_{low}, while $1 - \omega_{\mathrm{gate}}$ adjusts the weight for Fw_{high}. The weighted feature maps are then summed to obtain the fused feature map $Fefuse$. This process can be represented as

$$\begin{aligned} Fw &= Fw_{\mathrm{low}} + Fw_{\mathrm{high}} \\ \omega_{\mathrm{gate}} &= \sigma\left(\mathrm{MLP}(\mathrm{MP}(Fw)) + \mathrm{MLP}(\mathrm{AP}(Fw))\right) \\ Fe_{\mathrm{fuse}} &= \omega_{\mathrm{gate}} \times Fw_{\mathrm{low}} + \left(1 - \omega_{\mathrm{gate}}\right) \times Fw_{\mathrm{high}} \end{aligned} \tag{13}$$

where $\mathrm{MP}(\cdot)$ and $\mathrm{AP}(\cdot)$ represent max pooling and average pooling, respectively.

As shown in Fig. 2, our FGFM is applied for both adjacent-level and cross-level fusion process. The former involves transferring features progressively across adjacent levels, ensuring effective feature alignment and propagation from deep to shallow. To further refine feature representation and stabilize its numerical distribution, each FGFM is followed by a CBR convolution block with a 3×3 kernel size. Subsequently, FGFM performs cross-level fusion by aligning and integrating refined features from different levels with the highest resolution feature generated through adjacent-level propagation. Finally, the aligned fine-grained features from all four levels are summed to generate a more comprehensive change output.

F. Loss Function

Class imbalance is a common challenge in DL, particularly in CD tasks. This imbalance typically manifests as a majority of unchanged samples and a minority of changed samples. As a result, the network may exhibit a bias toward predicting the more frequent class, leading to reduced performance in detecting the minority class. To address this challenge, we employ a hybrid optimization approach that integrates both binary cross-entropy (BCE) loss and dice coefficient (Dice) loss. The BCE loss helps the network better fit binary data, while the Dice loss mitigates the effects of class imbalance. The formulations for these two losses are as follows:

$$\begin{aligned} L_{\mathrm{BCE}}\left(\hat{p}, p\right) &= -\left(\hat{p}\log(p) + \left(1 - \hat{p}\right)\log(1 - p)\right) \\ L_{\mathrm{Dice}}\left(\hat{p}, p\right) &= 1 - \frac{1 + 2\hat{p}p}{1 + \hat{p} + p} \end{aligned} \tag{14}$$

where $\hat{p}$ denotes the ground truth labels, while p denotes the predicted probability of the changed or unchanged.

At each feature level, this hybrid loss is applied, and the total loss is obtained by summing the losses across all four levels. This approach reinforces the constraints on feature learning. The total training loss is formulated as

$$L_{\mathrm{total}}\left(\hat{p}, p\right) = \sum_{i=1}^{4} \lambda_1 L_{\mathrm{BCE}}\left(\hat{p}, p\right) + \lambda_2 L_{\mathrm{Dice}}\left(\hat{p}, p\right) \tag{15}$$

where λ_1 and λ_2 balance the two losses, and their values greatly affect model performance. The optimal values will be determined in the following discussion.

IV. Experiment

This section presents three public datasets, implementation details, and evaluation metrics. Seventeen methods are then

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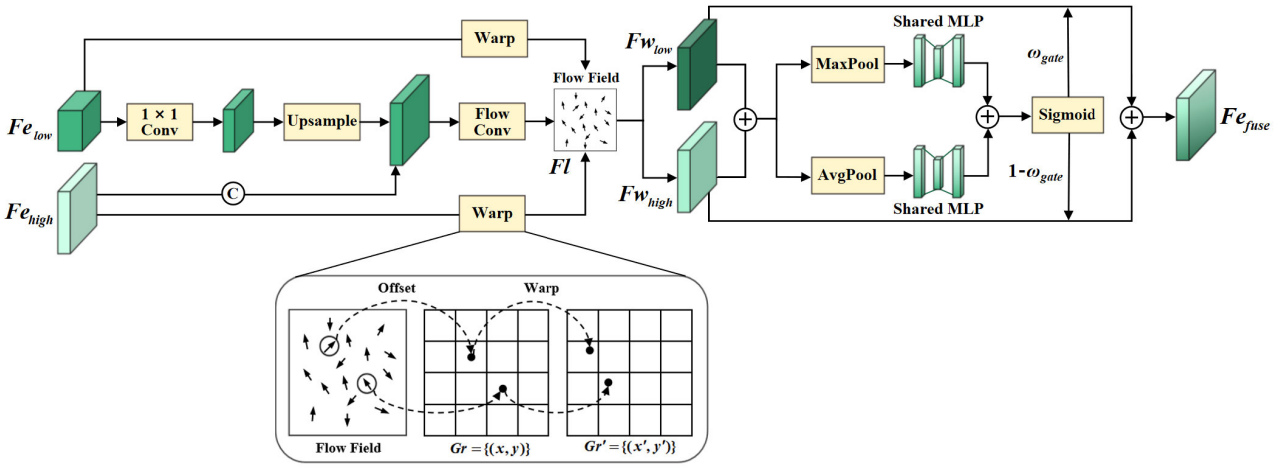


Fig. 7. Illustration of the FGFM.

selected for comparative experiments, followed by ablation experiments on different module combinations.

A. Datasets

Three openly accessible RS CD datasets are employed for training, validation, and testing: LEVIR-CD [30], WHU-CD [71], and SYSU-CD [72].

1) LEVIR-CD: This dataset offers a comprehensive collection of 637 pairs of bi-temporal, very high-resolution Google Earth image patches specifically designed for RS CD. Each patch has a resolution of 0.5 m and dimensions of 1024 × 1024 pixels, capturing land-use changes over periods of 5–14 years, focusing on building alterations. The dataset includes standard partitions for training, validation, and testing, each accompanied by corresponding binary labels. To enhance computational efficiency and reduce memory consumption, we segment the image pairs into nonoverlapping patches of 256 × 256 pixels. This segmentation results in 7120 image patch pairs for training, 1024 pairs for validation, and 2048 pairs for testing.

2) WHU-CD: This dataset comprises two collections of aerial imagery captured in 2012 and 2016, along with corresponding change maps specifically designed for building CD tasks. The RS images have a spatial resolution of 0.075 m and dimensions of 15 354 × 32 507 pixels. The original images are initially divided into smaller segments of 256 × 256 pixels, which are then randomly allocated to training, validation, and test sets in an 8:1:1 ratio. This allocation results in 5946 sample pairs for training, 744 pairs for validation, and 744 pairs for testing.

3) SYSU-CD: This dataset, released by Sun Yat-sen University, represents a substantial advancement over existing CD datasets in terms of image resolution, variety of change types, and dataset size. It comprises 20 000 pairs of 0.5-m resolution aerial images captured in Hong Kong between 2007 and 2014, along with corresponding binary change maps. Each image pair is 256 × 256 pixels in size. Following standard partitioning practices, the dataset is divided into 12 000 pairs for training, and 4000 pairs each for validation and testing.

B. Implementation Details and Evaluation Metrics

We construct the proposed network based on the PyTorch framework and perform training and testing on a single NVIDIA GeForce RTX 4090D GPU paired with an Intel[1] Xeon[1] Platinum 8747C CPU. The specific experimental setup includes Ubuntu 22.04, Python 3.10, PyTorch 2.4.0, and CUDA 11.8. The Adam optimizer [73] is selected with a momentum of 0.9, weight decay set to 0.0001, and $\beta 1$ and $\beta 2$ set to 0.9 and 0.99, respectively. The initial learning rate is set to $2e^{-4}$, and a poly learning adjustment scheme from [29] is applied. During training, the batch size is 16, and the model is trained for 100 epochs on three datasets. To enhance the model's robustness and generalization, data augmentation techniques such as random flipping, scaling, and cropping are applied during the preprocessing stage.

To quantitatively evaluate network performances, five indicators are employed: overall accuracy (OA), intersection over union (IoU), precision (P), recall (R), and F1-score (F1). These five measures are described as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{16}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \tag{17}$$

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{18}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{19}$$

$$\mathrm{F1} = \frac{2\times \mathrm{P}\times \mathrm{R}}{\mathrm{P}+\mathrm{R}} \tag{20}$$

where TP and FP denote the true positives and false positives, respectively, while TN and FN represent the true negatives and false negatives.

C. Compared Methods

To comprehensively evaluate the effectiveness of our network, we compare it with 17 representative DL-based CD methods. These include FC-Siam-diff [17], IFNet [26],

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SNUNet [18], HANet [32], USSFC-Net [31], EGRCNN [37], EGPNet [39], AERNet [28], P2V-CD [74], CGNet [75], BIT [43], ChangeFormer [44], IDA-SiamNet [64], ChangerEx(MiT-b0) [62], ChangeViT-S [50], MambaBCD-Tiny [52], and MambaBCD-Small [52]. These methods are categorized into three groups: CNN-based (C), transformer-based (T), and Mamba-based (M) models. To ensure fairness, the data augmentation techniques are also applied to all these models.

1) FC-Siam-diff: A Siamese network that identifies changes by computing simple feature differences between bi-temporal RS images.

2) IFNet: A deeply supervised image fusion network that utilizes the first ten layers of VGG16 for feature extraction, integrated with a differential discrimination network to achieve precise change mapping.

3) SNUNet: A densely connected Siamese network featuring a UNet++ architecture and an ensemble CAM to enhance CD performance.

4) HANet: A network combining progressive foreground-balanced sampling (PFBS) and a lightweight hierarchical attention mechanism, where PFBS focuses on learning change features and the attention mechanism integrates multiscale features to refine spatial and temporal changes.

5) USSFC-Net: An ultralightweight network employing multiscale decoupled convolutions and spatial-spectral feature cooperation to enhance feature extraction while minimizing computational cost and model parameters.

6) EGRCNN: An edge-guided recurrent CNN that integrates discriminative features with edge structure priors, significantly improving the results of building boundary detection.

7) EGPNet: An edge-guided parallel network incorporating an edge-aware module and an edge-guidance feature module within a parallel encoding framework to improve edge representation and enhance CD performance.

8) AERNet: An attention-guided network designed for building CD, which improves feature extraction and boundary identification through global context aggregation and edge refinement.

9) P2V-CD: A pseudo-transition video-based network that constructs a pseudo-transition video to incorporate rich temporal information, using decoupled encoders for spatiotemporal modeling to improve CD performance.

10) CGNet: A change-guided network, which generates change maps from deep features and utilizes a self-attention module to enhance the expression of change features, addressing the issues of inaccurate edge detection and internal holes in the U-Net structure.

11) BIT: A hybrid CNN-transformer network combining spatial attention and Siamese structures to improve pixel-level CD results.

12) ChangeFormer: An innovative Siamese network that integrates transformer for feature extraction with an MLP to fuse multilevel differential features, ultimately generating change predictions.

13) IDA-SiamNet: An interactive and dynamic-aware Siamese network that tackles the challenges of varying building shapes and sizes by incorporating spatial and channel exchange feature interaction modules and a dynamic-deformable dual-alignment fusion module.

14) ChangerEx(MiT-b0): A transformer-based interactive network that employs MiT-b0 [76] as the backbone, integrating alternating interaction layers and a flow dual-alignment fusion module to improve the interaction and alignment of bi-temporal features.

15) ChangeViT-S: A network based on ViT that enhances large-scale CD by employing ViT-S [48] backbone, along with a detail-capture module and feature injector for capturing fine-grained information across multiple scales.

16) MambaBCD-Tiny: A binary CD network based on the VMamba-T [77] backbone, which extracts multitemporal image features using a Siamese encoder and accurately predicts change maps by incorporating a spatiotemporal relationship modeling mechanism in the change decoder.

17) MambaBCD-Small: A network similar to MambaBCD-Tiny, but with the backbone replaced by the VMamba-S [77].

D. Comparative Experiments

1) Experiments on the LEVIR-CD Dataset: The test results on the LEVIR-CD dataset are detailed in Table II, which provides a quantitative comparison of methods. The best values are highlighted in red, and the second-best values are highlighted in blue. While each method performs well due to the superior quality of the dataset, our proposed network still stands out. It surpasses the second-best method (IDA-SiamNet) with improvements of 0.07% in IoU, 0.04% in F1, and 1.47% in R. Although the improvements are modest, our method significantly surpasses others in key metrics such as OA, IoU, and F1. Moreover, USSFC-Net achieves the highest R-value, highlighting its advantage in reducing missed detections, while IDA-SiamNet demonstrates superiority in minimizing false positives with the highest P value.

Fig. 8 presents the qualitative comparison results on this dataset. The first group of change samples involves complex edge shapes, where our method demonstrates superior performance in reducing false negatives compared to others. The corresponding edge comparison, shown in the first group of Fig. 9, indicates that while most comparative methods maintain the basic shape, our method excels in correctness and completeness of the outermost edge prediction. The second group of samples involves CD under shadowed conditions with complex and indistinct edges. Results show that our method achieves the lowest miss and false detection rates, with outputs closest to the ground truth. The third group samples reflect weak-edge small object changes under limited lighting conditions, with edge results illustrated in the second group of Fig. 9. Comparative methods generally suffer from issues such as missed detections, false detections, and edge adhesion, whereas our method effectively captures subtle change areas and preserves details, demonstrating strong robustness. The fourth group of samples features spectral inconsistency. In this scenario, our method significantly reduces missed detections and achieves the best performance in terms of edge continuity and completeness. The fifth group also deals with spectral inconsistency and small-object changes, where no method fully detects the change regions. However, our method

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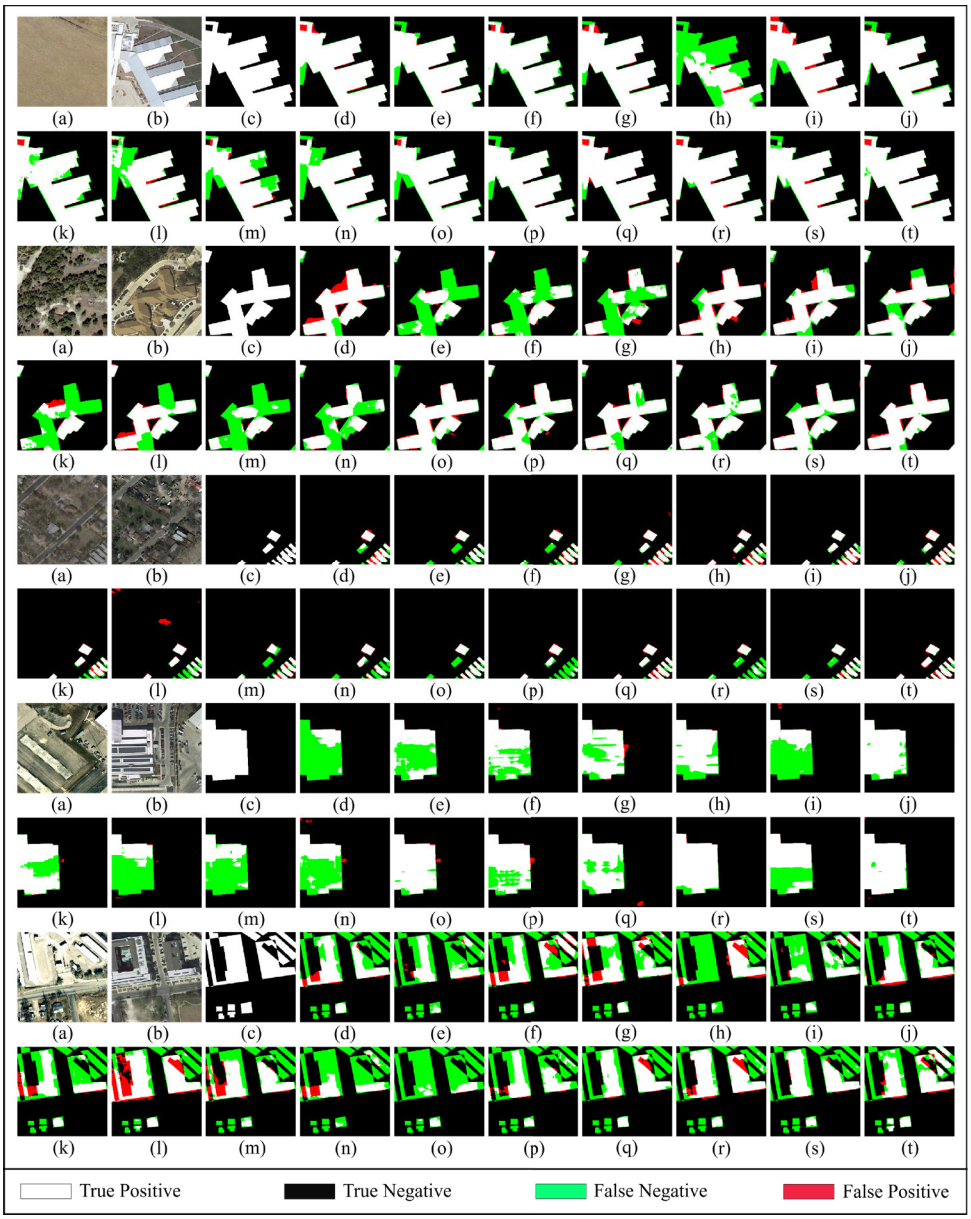


Fig. 8. Qualitative comparison results on LEVIR-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) IFNet. (e) SNUNet. (f) HANet. (g) USSFC-Net. (h) EGRCNN. (i) EGPNet. (j) AERNet. (k) P2V-CD. (l) CGNet. (m) BIT. (n) ChangeFormer. (o) IDA-SiamNet. (p) ChangerEx(MiT-b0). (q) ChangeViT-S. (r) MambaBCD-Tiny. (s) MambaBCD-Small. (t) Our EFICNN.

captures most of the change information through iterative learning and feature enhancement.

2) Experiments on the WHU-CD Dataset: Table II summarizes the quantitative comparison results of various methods on the WHU-CD dataset, with the highest values marked in red and the second-highest in blue. Our method outperforms the second-best method (IDA-SiamNet) by 0.41% in IoU, 0.22% in F1, and 0.73% in R. Although IDA-SiamNet achieves the highest P value, our method significantly surpasses all other methods across the remaining four metrics. These improvements highlight the superiority of our EFICNN.

Fig. 10 presents qualitative visualization results. The first group of change samples involves the addition and reduction of buildings at two time points. All comparison methods exhibit varying degrees of missed detections, with SNUNet failing to detect most of the changes. In contrast, our method is able to comprehensively detect internal changes while maintaining the integrity of the external edges. The second group of samples involves CD under different spectral conditions, where our EFICNN significantly reduces both false negatives and false positives. The edge changes correspond to the third group shown in Fig. 9, where other comparison methods fail to accurately depict the building shape, while our method's prediction closely matches the actual shape. The third group involves building changes with shadows and occlusions. IFNet and ChangeViT-S detect very few changes, whereas our EFICNN identifies a larger extent of real change regions. The fourth group of samples is affected by shadows and pseudo-changes, with edge predictions shown in the fourth group of Fig. 9. All comparison methods in this group demonstrate varying degrees of missed detections, false detections, or edge adhesion issues. In contrast, our method enhances the expression of

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THREE DATASETS

Models	Types	Params (M)	FLOPs (G)	LEVIR-CD OA/IoU/F1/R/P (%)	WHU-CD OA/IoU/F1/R/P (%)	SYSU-CD OA/IoU/F1/R/P (%)
FC-Siam-diff	*C*	**1.35**	**4.73**	98.72/76.99/87.00/83.94/90.29	98.26/67.70/80.74/73.93/88.93	85.66/41.62/58.78/43.34/**91.30**
IFNet		50.44	82.26	99.08/83.46/90.99/91.08/90.90	99.42/88.70/94.01/92.57/95.50	91.55/67.92/80.90/75.88/86.63
SNUNet		12.03	54.83	98.93/80.82/89.39/88.75/90.04	98.64/75.68/86.16/85.93/86.38	90.14/64.28/78.26/75.28/81.48
HANet		2.61	17.67	99.01/82.01/90.12/88.99/91.27	99.09/82.86/90.63/88.90/92.42	88.93/62.11/76.63/76.97/76.29
USSFC-Net		**1.52**	**4.86**	99.03/82.36/90.33/**92.01**/88.71	99.17/84.14/91.39/**93.72**/89.16	90.47/66.22/79.68/80.12/79.24
EGRCNN		9.63	17.64	98.90/80.82/89.39/90.60/88.21	99.15/83.58/91.05/88.14/94.16	90.60/66.89/80.16/80.50/79.82
EGPNet		44.32	77.33	99.04/82.42/90.36/88.78/92.01	99.25/85.45/92.15/88.83/95.74	91.38/67.90/80.88/77.30/84.82
AERNet		25.36	12.82	99.09/83.36/90.93/89.77/92.12	99.47/89.54/94.48/92.23/96.84	91.97/68.80/81.52/75.10/**89.13**
P2V-CD		5.42	32.96	**99.17**/84.77/91.75/90.40/93.15	99.23/85.08/91.94/88.82/95.29	90.40/64.43/78.37/73.77/83.58
CGNet		33.68	82.23	99.15/84.43/91.56/90.74/92.39	99.42/88.78/94.06/93.26/94.86	90.74/66.17/79.64/76.77/82.73
BIT	*T*	3.04	8.75	98.98/81.72/89.94/89.56/90.33	98.93/80.17/88.99/87.41/90.64	89.89/64.33/78.29/77.30/79.31
ChangeFormer		41.03	202.79	99.00/81.88/90.04/88.07/92.10	98.99/80.47/89.18/84.50/94.40	90.52/65.75/79.34/77.17/81.63
IDA-SiamNet		14.14	25.21	**99.19**/**85.09**/**91.94**/90.18/**93.78**	**99.55**/**91.21**/**95.41**/93.68/**97.20**	**92.16**/70.09/82.42/77.96/87.41
ChangerEx(MiT-b0)		3.45	8.52	99.15/84.37/91.52/89.75/**93.37**	99.27/85.80/92.36/89.31/95.61	91.99/70.00/82.35/79.21/85.75
ChangeViT-S		32.14	38.81	99.16/84.68/91.70/91.28/92.13	99.44/89.11/94.24/92.96/95.56	91.67/69.77/82.20/81.53/82.87
MambaBCD-Tiny	*M*	17.13	45.74	99.10/83.72/91.14/90.68/91.60	99.50/90.24/94.87/93.44/96.35	91.66/68.79/81.51/77.99/85.37
MambaBCD-Small		49.94	114.82	99.11/83.74/91.15/90.39/91.93	99.49/90.07/94.78/92.96/96.66	91.92/**70.42**/**82.64**/**81.51**/83.80
EFICNN(Ours)	*C*	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	**99.57**/**91.62**/**95.63**/**94.41**/**96.87**	**92.13**/**70.97**/**83.02**/**81.63**/84.46

change subjects and boundaries through edge optimization and feature enhancement. The fifth group involves changes with blurred edges under spectral inconsistency, where P2V-CD shows significant missed detections, while other comparison methods detect notable pseudo changes. Although AERNet and MambaCD-Small perform relatively well among the comparison methods, our prediction is closer to the ground truth.

3) Experiments on the SYSU-CD Dataset: Table II shows the best values of IoU, F1, and *R* evaluation metrics achieved by our method on the SYSU-CD dataset. Compared to the second-best method (MambaBCD-Small), our method improves IoU, F1, *R*, and *P* by 0.55%, 0.38%, 0.12%, and 0.66%, respectively. In contrast, FC-Siam-diff performs the worst due to inherent information loss from its simple feature differencing operation. This method shows the highest *P* but low *R*, indicating that it accurately identifies some changed areas while missing a significant number of actual changes. Additionally, the highest OA achieved by IDA-SiamNet indicates its outstanding performance in classifying change and nonchange regions, while the OA of our EFICNN is second only to it, showing strong competitiveness.

Compared to other methods, our approach demonstrates superior visual performance, as shown in Fig. 11. The first group of samples presents irregular bare land changes, where all comparison methods exhibit significant false negatives, with some methods also showing false positives at the edges. In contrast, our method best detects the change regions and their edges. In the second group of samples, our method clearly detects changes in ships while maintaining the lowest false detection rate along the port edge. The third group presents road changes, with the output edges corresponding to the fifth group results in Fig. 9. It can be observed that all comparison methods exhibit significant false negatives. In contrast, our method more accurately detects straight road lines while preserving the internal integrity. The fourth group presents building changes, where the external edges are more blurred. Although EGPNet, ChangeViT-S, MambaBCD-Tiny, and MambaBCD-Small perform better in comparison, our EFICNN more accurately reflects the actual changes. The fifth group samples in Fig. 11 present new bare land changes with blurred edges, with the corresponding edge prediction results shown in the sixth group of Fig. 9. While all comparison methods detect the basic shape of the bare land, most methods exhibit varying degrees of missed or false detections. Although BIT, EGRCNN, and CGNet perform well, the binary output and edge produced by our method are closest to the ground truth label.

E. Efficiency Comparisons

The efficiency comparison is conducted using the torchstat library in PyTorch, which calculates the model's parameter count (Params) and floating-point operations (FLOPs) to evaluate its computational cost and speed. Table II presents the efficiency results of various methods. The data reveals that although FC-Siam-diff has the lowest Params and FLOPs, its quantitative results are considerably inferior to other models across three datasets. This is primarily due to the limited number of convolutional layers, which restricts its feature extraction capability. Lightweight models such as HANet, USSFC-Net, BIT, and ChangerEx (MiT-b0) also have few Params and FLOPs but fail to deliver exceptional performance on all datasets simultaneously, indicating their limitations in handling diverse tasks. Conversely, models with larger Params and FLOPs, such as IFNet, ChangeFormer, EGPNet, CGNet, and MambaBCD-Small, generally improve feature extraction and CD results by utilizing more complex networks or larger convolutions. Despite the increased computational complexity, these models still fall short of outperforming our model on three datasets. In contrast, our EFICNN exhibits

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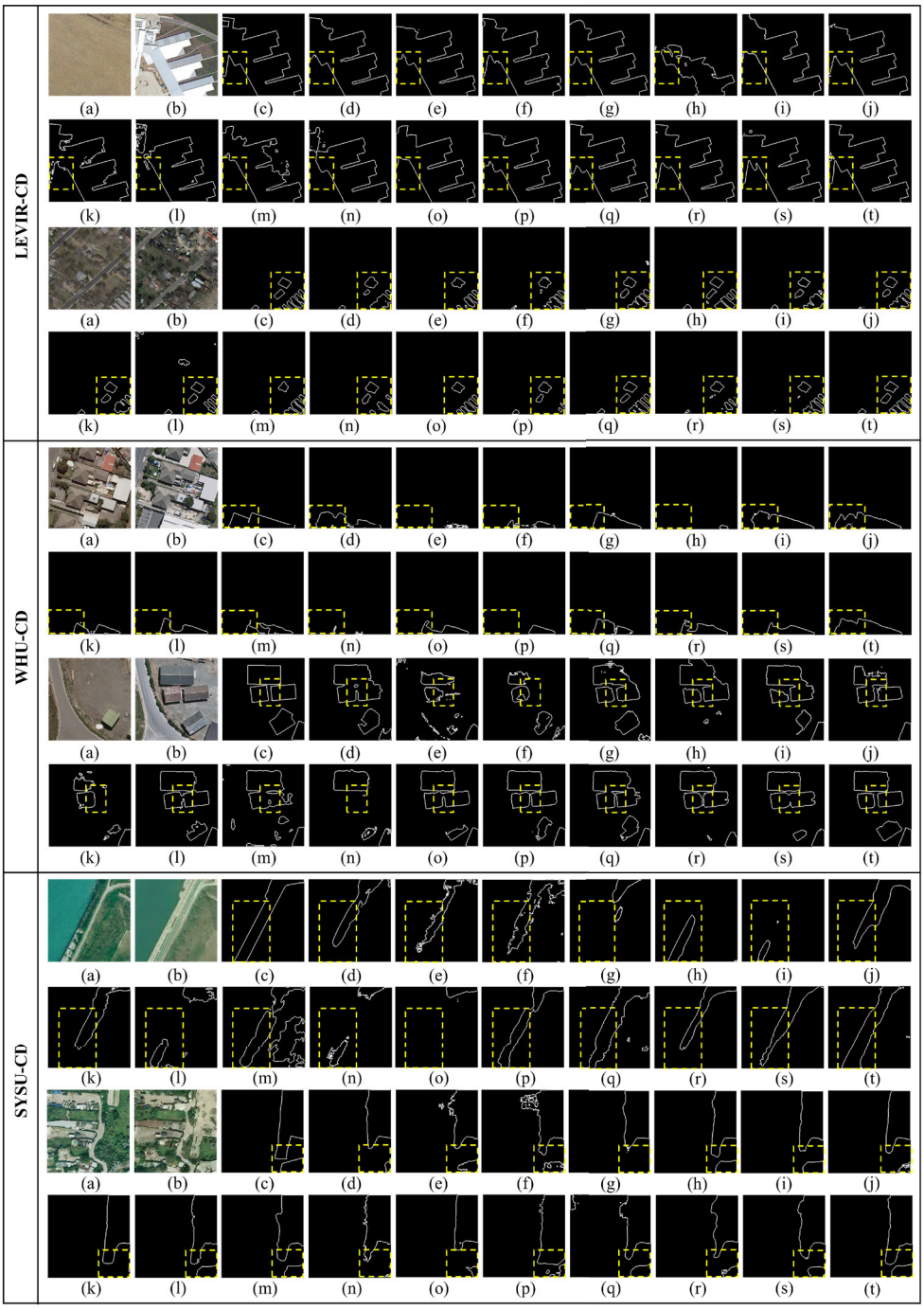


Fig. 9. Edge comparison results on three dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) IFNet. (e) SNUNet. (f) HANet. (g) USSFC-Net. (h) EGRCNN. (i) EGPNet. (j) AERNet. (k) P2V-CD. (l) CGNet. (m) BIT. (n) ChangeFormer. (o) IDA-SiamNet. (p) ChangerEx(MiT-b0). (q) ChangeViT-S. (r) MambaBCD-Tiny. (s) MambaBCD-Small. (t) Our EFICNN.

moderate efficiency and achieves the optimal balance between computational efficiency and performance, attaining SOTA results among 17 popular models.

F. Ablation Experiments

To validate the effectiveness of the IB, 3D-DEM, EGAM, and FGFM, we conduct ablation experiments on three datasets. The quantitative results are shown in Table III. The baseline model (#01) uses a single VGG16 to extract bi-temporal features, which are then processed through a simple difference module (DM) in Fig. 12 to generate the change maps. This baseline does not optimize edge information, but directly uses bilinear interpolation for feature upsampling and pixel-wise addition to fuse maps at different levels for the final output.

In the single-module experiments, adding IB (#02), 3D-DEM (#03), EGAM (#04), and FGFM (#05) all improve CD results across three datasets. Among these, IB shows the most significant improvement by iteratively enhancing bi-temporal features, while FGFM notably improves feature fusion through optical flow alignment and GM. The 3D-DEM

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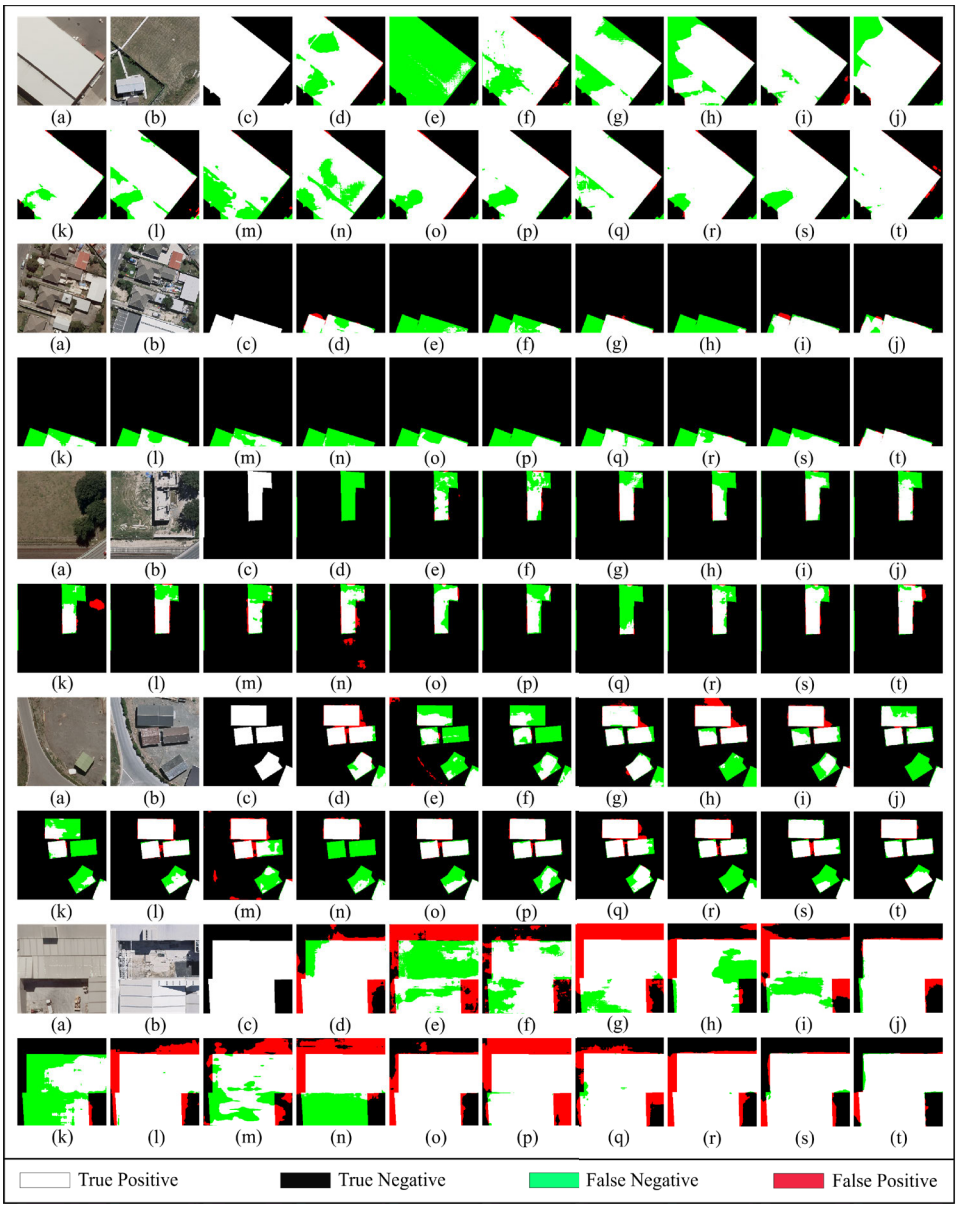


Fig. 10. Qualitative comparison results on WHU-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) IFNet. (e) SNUNet. (f) HANet. (g) USSFC-Net. (h) EGRCNN. (i) EGPNet. (j) AERNet. (k) P2V-CD. (l) CGNet. (m) BIT. (n) ChangeFormer. (o) IDA-SiamNet. (p) ChangerEx(MiT-b0). (q) ChangeViT-S. (r) MambaBCD-Tiny. (s) MambaBCD-Small. (t) Our EFICNN.

and EGAM also contribute to performance gains, especially on the SYSU-CD dataset, highlighting their strength in handling large-scale samples and complex change scenarios. In dual-module experiments, the combination of IB and FGFM (#09) further boosts performance, particularly on the LEVIR-CD and SYSU-CD datasets, with a trend consistent with the notable gains observed when using IB and FGFM individually. On the WHU-CD dataset, the IB and 3D-DEM combination (#06) outperforms others, highlighting the complementary advantages of these two modules. For the three-module combinations, the IB, EGAM, and FGFM combination (#13) achieves the best results on LEVIR-CD and WHU-CD, while IB, 3D-DEM, and FGFM (#14) perform best on SYSU-CD. This highlights the advantages of IB and FGFM, as well as the varying contributions of 3D-DEM and EGAM under different combination modes. When all four modules are combined (#16), the model performance reaches its peak, with IoU and F1 increasing by 2.87% and 1.70% on LEVIR-CD, 3.93% and 2.19% on WHU-CD, and 6.62% and 4.71% on SYSU-CD compared to the baseline. Additionally, by progressively adding these four modules to the baseline model, it is observed that the model's IoU and F1 steadily improve across three datasets, demonstrating the effectiveness of each module.

Moreover, Fig. 13(a) and (b), respectively, illustrates the F1 and loss on the validation set for each epoch. It can be observed that the progressive integration of these four modules leads to a steady decrease in model loss and a corresponding increase in F1, demonstrating their positive impact on CD performance. Moreover, a comparative analysis of different module combinations indicates that EFICNN maintains a more stable validation curve, with F1 consistently remaining at a relatively high level. These findings confirm the robustness and superiority of our model.

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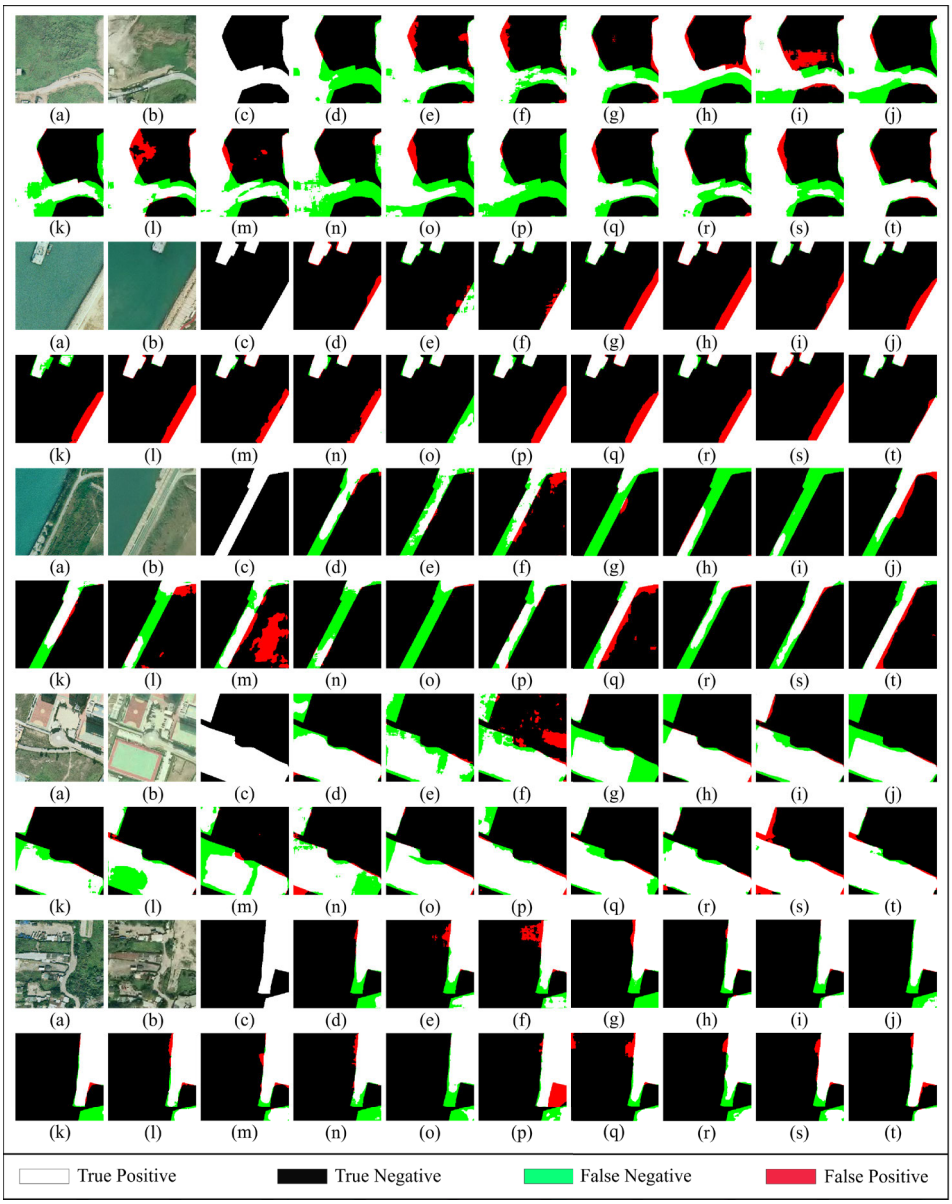


Fig. 11. Qualitative comparison results on SYSU-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) IFNet. (e) SNUNet. (f) HANet. (g) USSFC-Net. (h) EGRCNN. (i) EGPNet. (j) AERNet. (k) P2V-CD. (l) CGNet. (m) BIT. (n) ChangeFormer. (o) IDA-SiamNet. (p) ChangerEx(MiT-b0). (q) ChangeViT-S. (r) MambaBCD-Tiny. (s) MambaBCD-Small. (t) Our EFICNN.

Qualitative ablation experimental results are shown in Fig. 14. The baseline model exhibits the highest proportion of false negatives or false positives across three datasets. With the incorporation of IB, the model captures more change-related features, leading to a noticeable expansion of the detected change areas. Building on this, the addition of the 3D-DEM obviously suppresses pseudo changes while enhancing the saliency of real change regions. The introduction of EGAM further enhances the delineation of change edges, refining their smoothness and continuity. Finally, the FGFM facilitates the fusion of multiscale features, ensuring better preservation of key change regions. Overall, the progressive integration of these modules effectively enhances the performance of CD.

To more clearly demonstrate the effectiveness of each module, we visualize the output probability maps of four-level features at different stages. As shown in Fig. 15(a), the RGB inputs of two time points and their corresponding probability maps are presented. After feature extraction through the IB, the outputs are shown in Fig. 15(b) and (c). Compared to Fig. 15(a), the main objects are more prominently emphasized, with the shallow features preserving abundant spatial details and the deep features highlighting critical semantic information. Subsequently, the difference probability maps generated by the 3D-DEM are shown in Fig. 15(d). Compared to Fig. 15(b) and (c), these maps more clearly reflect the locations and contours of changes. Although the contours are not entirely well-defined, this module has effectively detected real changes while suppressing pseudo changes. Next, the EGAM further enhances the clarity of change regions through edge enhancement, reducing the background noise and emphasizing the edge details, as illustrated in Fig. 15(e). Subsequently, the FGFM aligns and fuses low-level detail

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TABLE III
QUANTITATIVE ABLATION EXPERIMENTAL RESULTS ON THREE DATASETS

No.	Models				Params	FLOPs	LEVIR-CD	WHU-CD	SYSU-CD
	IB	3D-DEM	EGAM	FGFM	(M)	(G)	OA/IoU/F1/R/P (%)	OA/IoU/F1/R/P (%)	OA/IoU/F1/R/P (%)
#01	×	×	×	×	7.64	18.28	99.02/82.29/90.28/89.74/90.83	99.36/87.69/93.44/92.06/94.86	90.32/64.35/78.31/74.13/82.99
#02	✓	×	×	×	10.94	20.80	99.12/83.74/91.15/88.97/**93.43**	99.43/88.90/94.12/92.08/96.26	90.94/66.01/79.52/74.62/85.12
#03	×	✓	×	×	13.90	23.12	99.06/82.98/90.70/89.92/91.49	99.42/88.53/93.92/91.32/96.66	90.66/65.24/78.97/74.31/84.24
#04	×	×	✓	×	8.34	18.83	99.05/82.81/90.59/89.98/91.22	99.42/88.70/94.01/92.90/95.15	90.79/65.36/79.05/73.71/**85.23**
#05	×	×	×	✓	11.05	21.11	99.10/83.63/91.08/89.92/92.27	99.41/88.49/93.89/92.00/95.86	90.41/65.88/79.43/78.53/80.35
#06	✓	✓	×	×	17.21	25.64	99.13/84.08/91.35/90.06/92.68	99.46/89.40/94.40/91.86/97.09	90.69/66.53/79.90/78.50/81.36
#07	✓	×	✓	×	11.65	21.35	99.13/84.01/91.31/89.77/92.90	99.45/89.28/94.33/92.40/96.35	90.93/66.35/79.77/75.80/84.18
#08	✓	×	×	✓	14.35	23.63	99.14/84.36/91.52/91.60/91.44	99.43/88.94/94.15/92.94/95.38	91.59/68.80/81.51/78.60/84.65
#09	×	✓	✓	×	14.61	23.67	99.11/83.64/91.09/89.75/92.47	99.43/88.97/94.16/92.87/95.50	90.86/65.97/79.50/75.11/84.43
#10	×	✓	×	✓	17.32	25.95	99.12/83.86/91.22/89.46/93.05	99.43/88.98/94.17/93.57/94.77	90.82/66.53/79.90/77.38/82.59
#11	×	×	✓	✓	11.75	21.66	99.12/83.96/91.28/90.71/91.87	99.44/89.10/94.24/93.15/95.35	91.05/66.59/79.95/75.68/84.73
#12	×	✓	✓	✓	18.02	26.50	99.14/84.15/91.39/89.70/93.15	99.45/89.33/94.37/93.25/95.51	90.91/66.96/80.21/78.15/82.39
#13	✓	×	✓	✓	15.06	24.18	99.16/84.70/91.72/91.35/92.08	99.54/91.05/95.32/94.14/96.53	91.55/69.26/81.84/80.78/82.92
#14	✓	✓	×	✓	20.62	28.47	99.15/84.56/91.63/**91.77**/91.50	99.54/90.99/95.28/93.86/96.75	91.75/69.63/82.10/80.23/84.06
#15	✓	✓	✓	×	17.91	26.19	99.14/84.34/91.50/90.53/92.50	99.49/89.89/94.68/92.39/**97.09**	90.83/67.69/80.74/81.47/80.01
#16	✓	✓	✓	✓	21.32	29.02	**99.19**/**85.16**/**91.98**/91.65/92.32	**99.57**/**91.62**/**95.63**/**94.41**/96.87	**92.13**/**70.97**/**83.02**/**81.63**/84.46

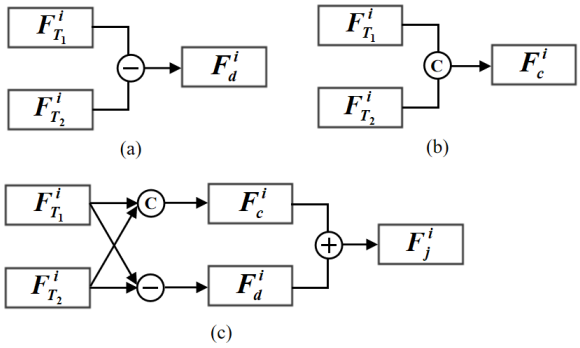


Fig. 12. Schematic of distinct DMs. (a) DM. (b) CM. (c) Combining DCM. Here, F_c^i represents the concatenated feature, F_d^i denotes the difference feature, and F_j^i signifies the joint feature.

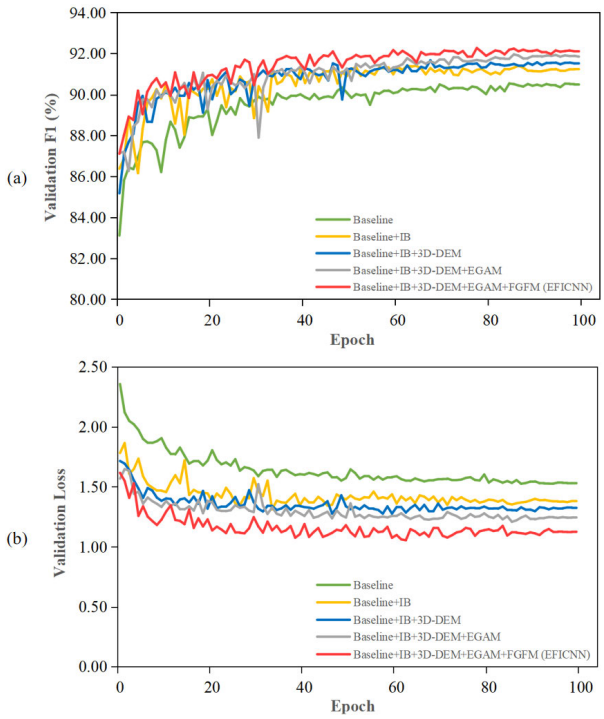


Fig. 13. Visualization of validation F1 and loss during the training process on LEVIR-CD dataset. (a) Validation F1 curves. (b) Validation loss curves.

information with high-level semantic features, producing the probability maps shown in Fig. 15(f). The outputs from this module closely match the real changes, demonstrating precise target localization and clear edges. Finally, the four-level features from the FGFM are summed to generate the final prediction probability map, as presented in the first row of Fig. 15(g). This result highlights the primary change objects with the highest attention, achieving nearly perfect alignment with the ground truth label, thereby validating the effectiveness of our combined approach.

V. DISCUSSION

In this section, building upon the preliminary ablation experiments, we further provide an in-depth discussion within each module to explore the specific contributions of key components to performance improvements. Additionally, we expand the discussion to explore alternative variants of the modules, highlighting the advantages of the proposed design.

A. Discussion on the IB

In the ablation experiments, we have demonstrated the importance of the IB. To further investigate the specific contributions of its key components, we conduct ablation analysis on the iterative structure, FPN-like multiscale feature enhancement, and fusion operation. Quantitative results shown in Table IV (#01–#04), indicate that removing the multiscale feature enhancement has the most significant impact on performance across three datasets, particularly on WHU-CD and SYSU-CD datasets. This highlights the critical role of this component in feature perception. By effectively integrating multiscale feature maps, it significantly enhances the model's contextual understanding of changes. As illustrated in Fig. 16(e), removing it leads to a noticeable decline in the model's ability to detect complex shape changes, edge details, and small objects, while its capacity to suppress

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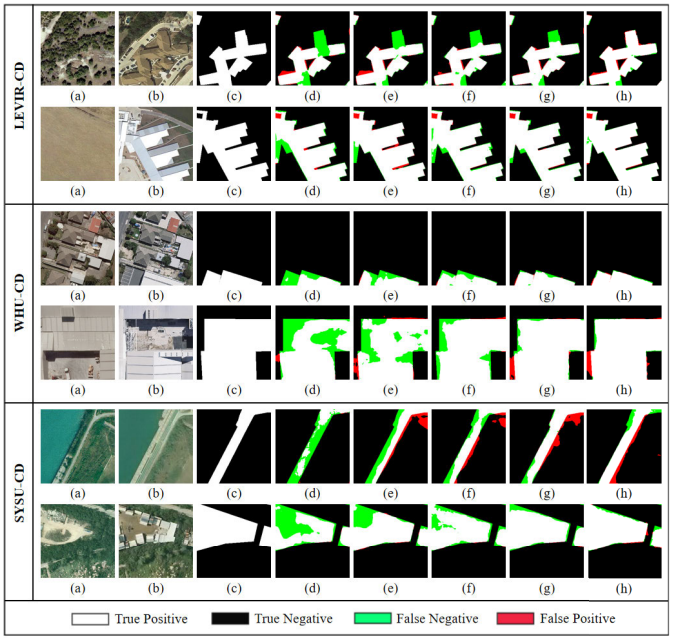


Fig. 14. Ablation experimental results on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) Baseline. (e) Baseline + IB. (f) Baseline + IB + 3D-DEM. (g) Baseline + IB + 3D-DEM + EGAM. (h) Baseline + IB + 3D-DEM + EGAM + FGFM (Our EFICNN).

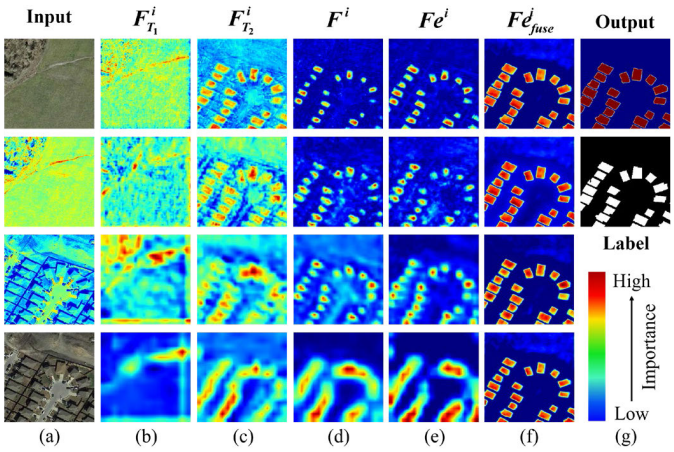


Fig. 15. Visualizations of four-level feature maps. (a) Bi-temporal RGB images and their probability maps. (b) First-temporal probability maps from the IB. (c) Second-temporal probability maps from the IB. (d) Probability maps from the 3D-DEM. (e) Probability maps from the EGAM. (f) Probability maps from the FGFM. (g) Final predicted probability map and ground truth label. In these visualizations, red represents significant change regions, while blue indicates areas of lower importance.

pseudo changes is also weakened. In addition, the iteration optimizes the composite feature extraction through a dual-backbone, while the fusion operation efficiently integrates the outputs of two backbones, further strengthening the perception of changes. The absence of these two components leads to varying degrees of metrics degradation, with corresponding visual outputs shown in Fig. 16(d) and (f). In these cases, the results exhibited more pronounced omissions and false detections, particularly in complex scenarios. These findings underscore the indispensable role of IB and its components in improving the overall performance and robustness of our model.

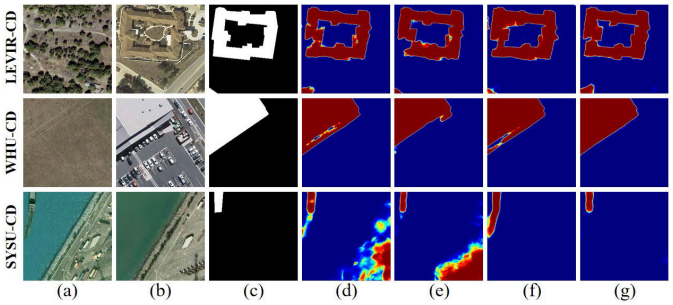


Fig. 16. Visualizations of the final predicted features on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) Without iteration. (e) Without FPN. (f) Without fusion. (g) Our EFICNN.

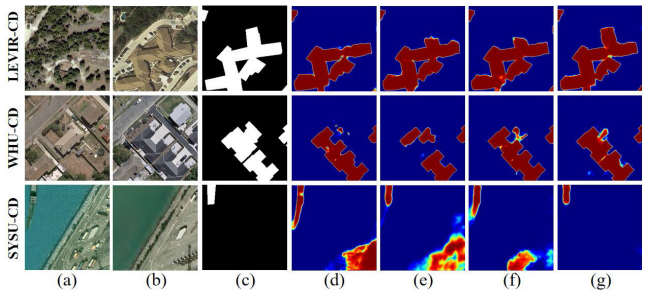


Fig. 17. Visualizations of the final predicted features on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) With DM. (e) With CM. (f) With DCM. (g) With our 3D-DEM.

To validate the rationality and flexibility of the iterative structure, we assess its performance using various backbone networks, including MobileNetV2 [22], ResNet18 [21], Swin-T [78], and VMamba-T [77]. These backbones represent CNN, transformer, and Mamba architectures, respectively. The quantitative comparison results, presented in Table IV (#05–#08), reveal distinct performance patterns across three datasets. Among the models, the iterative MobileNetV2 demonstrates the weakest performance. Its use of depthwise separable convolutions and lightweight design effectively reduces Params and FLOPs. However, these optimizations limit its feature representation capacity. In contrast, the iterative Swin-T excels in capturing long-term dependencies and global information, outperforming the iterative VGG16 on the SYSU-CD dataset. Nonetheless, its performance lags behind ours on the LEVIR-CD and WHU-CD datasets, which also reflects its higher computational requirements. Similarly, the iterative ResNet18 surpasses our iterative VGG16 on the SYSU-CD but exhibits inferior performance on the LEVIR-CD and WHU-CD datasets. This disparity likely arises from VGG16's ability to produce higher-resolution feature maps at the same number of channels, enabling it to capture more information. Notably, the iterative VMamba-T achieves the best performance across all three datasets, albeit at the cost of higher computational demands. Its global receptive field and dynamic weighting mechanisms empower VMamba-T with superior feature extraction capabilities. These results highlight the varying adaptability of backbones to different datasets and emphasize the rationality and flexibility of our model design.

To further confirm the scalability of the IB, we extend our discussion by introducing a third VGG16 alongside the dual backbones. The iterative connection between the three back-

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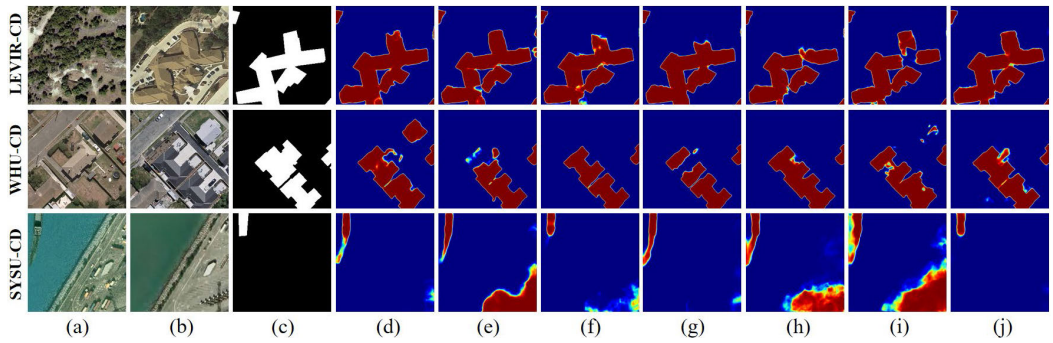


Fig. 18. Visualizations of the final predicted features on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) Without Laplacian pyramid. (e) Without reverse attention. (f) With isotropic Sober. (g) With Krisch. (h) With Laplacian. (i) With LoG. (j) Our EFICNN.

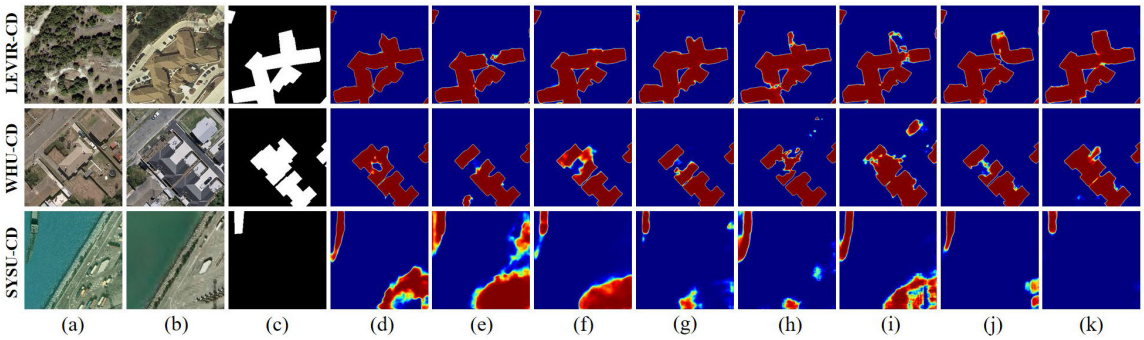


Fig. 19. Visualizations of the final predicted features on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) Without GM. (e) Without adjacent fusion. (f) Without cross fusion. (g) With FAM1 (bilinear interpolation). (h) WithFAM2 (unpooling). (i) WithFAM3 (deconvolution). (j) With FAM4 (two channel flow alignment). (k) Our EFICNN.

bones follows the same scheme as the dual-backbone network. To ensure comparability, the single VGG16 retains the multi-scale feature enhancement, which differs from the previous module ablation experiments. The quantitative results presented in Table IV (#09–#11), indicate that our dual VGG16 architecture significantly outperforms the single one across all evaluation metrics on three datasets. However, as the number of backbones increases, the evaluation metrics decline, particularly on the WHU-CD and SYSU-CD datasets. We speculate that the introduction of a third backbone may result in the overlap of features extracted from the initial two backbones, thereby affecting CD performance. Although parameter sharing between backbones does not increase Params and FLOPs, it introduces the risk of overfitting. These results thus confirm the effectiveness of our iterative dual VGG16 design.

B. Discussion on the 3D-DEM

The 3D-DEM combines difference and concatenation branches to capture more change information, while the SimAM attention mechanism highlights discriminative features. To further validate the effectiveness of this combination, we conduct experiments to separately test the impact of individual branches, the fused structure, and the attention mechanism. The quantitative results in Table IV (#12–#15) indicate that using either the concatenation module (CM) or DM alone is insufficient for capturing change information or preserving contextual details, leading to relatively lower performance. In contrast, the combination of these two modules difference and concatenation module (DCM) improves the metrics on three datasets. Moreover, the introduction of the SimAM attention mechanism in each branch further enhances performance by emphasizing discriminative features without adding extra parameters. Fig. 17(d)–(f) presents the visualization of the final predicted features for DM, CM, and DCM, while Fig. 17(g) shows the corresponding results for 3D-DEM. In these visualizations, red indicates areas with significant changes, and blue represents regions with minimal or insignificant changes. The results demonstrate that the outputs of CM and DM exhibit a considerable number of missed and false detections, whereas DCM combines the advantages of both and performs better. In contrast, the 3D-DEM effectively focuses on the real changed areas while minimizing the impact of noise and irrelevant regions.

C. Discussion on the EGAM

In the EGAM, the combination of reverse attention for background changes and edge attention based on the Laplacian pyramid aims to optimize the detection of change objects and enhance high-frequency edge information. As shown in Table IV (#16–#17), removing these two components obviously reduces overall performance. The corresponding visual results can be found in Fig. 18(d) and (e). When this operator is omitted and only the self-attention of the edges is calculated, there is a lack or over feature expressions in the edge areas. The absence of reverse attention results in the ineffective preservation of background changes, leading to more significant omissions and false detections.

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TABLE IV
QUANTITATIVE RESULTS OF KEY COMPONENT ABLATION ON THREE DATASETS

No.	Models	Params (M)	FLOPs (G)	LEVIR-CD OA/IoU/F1/R/P (%)	WHU-CD OA/IoU/F1/R/P (%)	SYSU-CD OA/IoU/F1/R/P (%)
(a) IB						
#01	w/o Iteration	21.32	29.02	99.16/84.68/91.71/91.49/91.93	99.54/91.07/95.33/**94.81**/95.84	91.62/70.16/82.46/**83.52**/81.43
#02	w/o FPN	18.02	26.50	99.15/84.45/91.57/90.53/92.64	99.29/86.41/92.71/91.18/94.29	91.56/68.79/81.51/78.87/84.32
#03	w/o Fusion	21.32	29.02	99.17/84.83/91.79/90.80/**92.81**	99.55/91.27/95.44/94.42/96.47	91.81/69.75/82.18/80.04/84.43
#04	**Our EFICNN**	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	**99.57**/**91.62**/**95.63**/94.41/**96.87**	**92.13**/**70.97**/**83.02**/81.63/**84.46**
(b) Different IB						
#05	MobileNetV2	6.22	2.45	99.07/83.18/90.82/90.35/91.30	99.48/89.93/94.70/93.32/96.12	91.95/70.11/82.43/80.10/84.90
#06	Swin-T	49.61	25.83	99.15/84.51/91.60/90.90/92.32	99.55/91.11/95.35/94.11/96.61	92.15/71.27/83.22/82.51/83.95
#07	Vmamba-T	56.75	31.47	**99.19**/**85.25**/**92.04**/91.43/92.65	**99.58**/**91.76**/**95.70**/**94.70**/96.72	**92.56**/**72.02**/**83.73**/81.22/**86.41**
#08	ResNet18	24.87	14.13	99.09/83.41/90.96/89.83/92.11	99.49/90.06/94.77/93.70/95.86	92.52/71.98/83.70/81.49/86.04
#09	Single VGG16	21.32	29.02	99.17/84.80/91.77/91.12/92.43	99.52/90.73/95.14/94.33/95.96	91.70/69.42/81.95/79.88/84.13
#10	**Our Dual VGG16**	21.32	29.02	99.19/85.16/91.98/**91.65**/92.32	99.57/91.62/95.63/94.41/**96.87**	92.13/70.97/83.02/81.63/84.46
#11	Triple VGG16	21.32	29.02	99.19/85.13/91.97/91.06/**92.89**	99.33/86.97/93.03/90.40/95.82	91.64/70.22/82.50/**83.60**/81.44
(c) 3D-DEM						
#12	w/ DM	15.06	24.18	99.16/84.70/91.72/91.35/92.08	99.54/91.05/95.32/94.14/96.53	91.55/69.26/81.84/80.78/82.92
#13	w/ CM	21.32	29.02	99.18/84.79/91.77/90.14/**93.47**	99.54/91.08/95.33/94.29/96.39	91.18/69.28/81.85/**84.34**/79.50
#14	w/ DCM	21.32	29.02	99.18/84.91/91.84/90.77/92.94	99.55/91.19/95.39/94.28/96.53	91.94/70.50/82.70/81.66/83.76
#15	**w/ Our 3D-DEM**	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	**99.57**/**91.62**/**95.63**/**94.41**/**96.87**	**92.13**/**70.97**/**83.02**/81.63/**84.46**
(d) EGAM						
#16	w/o Laplacian pyramid	21.32	29.02	99.17/84.80/91.77/90.80/92.77	99.48/89.90/94.68/93.69/95.70	91.78/70.30/82.56/82.55/82.57
#17	w/o Reverse attention	20.98	28.75	99.16/84.70/91.72/91.06/92.38	99.55/91.23/95.41/**94.97**/96.10	91.85/70.29/82.56/81.75/83.38
#18	w/ Isotropic Sober	21.32	29.02	99.18/84.99/91.88/90.68/**93.12**	99.56/91.34/95.47/94.46/96.51	91.47/70.11/82.43/84.88/80.12
#19	w/ Krisch	21.32	29.02	99.18/85.08/91.94/91.44/92.44	99.52/90.61/95.08/93.82/96.37	91.73/70.02/82.37/81.92/82.82
#20	w/ Laplacian	21.32	29.02	99.16/84.70/91.72/90.97/92.48	99.56/91.37/95.49/94.89/96.10	91.22/70.15/82.46/**87.50**/77.97
#21	w/ LoG	21.32	29.02	99.19/85.15/91.98/91.12/92.85	99.37/88.04/93.64/93.67/93.61	91.08/69.02/81.67/84.24/79.26
#22	**w/ Our Laplacian pyramid**	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	**99.57**/**91.62**/**95.63**/94.41/**96.87**	**92.13**/**70.97**/**83.02**/81.63/**84.46**
(f) FGFM						
#23	w/o GM	21.31	29.02	99.17/84.89/91.83/91.07/92.60	99.55/91.17/95.38/94.02/96.78	91.89/70.06/82.39/80.44/84.45
#24	w/o Adjacent fusion	21.12	28.89	99.18/84.90/91.83/90.66/93.03	99.56/91.42/95.52/94.13/96.95	91.78/70.10/82.42/80.78/84.13
#25	w/o Cross fusion	21.27	28.84	99.18/84.72/91.73/89.72/**93.38**	99.55/91.20/95.40/93.67/97.18	91.61/69.81/82.22/82.29/82.16
#26	w/ FAM1 (Interpolation)	21.28	28.74	99.18/84.95/91.86/90.69/93.07	99.54/90.95/95.26/93.81/96.75	91.75/70.72/82.85/**84.47**/81.30
#27	w/ FAM2 (Unpooling)	21.28	29.38	99.17/84.80/91.77/90.80/92.77	99.53/90.86/95.21/93.99/96.47	91.04/67.36/80.50/78.46/82.65
#28	w/ FAM3 (Deconvolution)	24.11	28.61	99.19/85.07/91.93/91.12/92.76	99.57/91.48/95.55/93.98/**97.18**	91.86/69.95/82.32/80.34/84.40
#29	w/ FAM4 (Two channel)	21.30	28.88	99.18/85.02/91.90/91.40/92.41	**99.58**/**91.74**/**95.69**/94.32/97.10	91.84/69.83/82.24/80.06/**84.54**
#30	**w/ Our FAM (Four channel)**	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	99.57/91.62/95.63/**94.41**/96.87	**92.13**/**70.97**/**83.02**/81.63/84.46
(g) Loss Function						
#31	BCE	21.32	29.02	99.14/84.23/91.44/89.75/**93.20**	99.55/91.18/95.39/94.40/96.40	91.91/**71.00**/**83.04**/83.98/82.12
#32	Dice	21.32	29.02	99.17/84.88/91.82/91.02/92.64	99.47/89.56/94.49/92.08/**97.04**	91.22/68.47/81.28/80.86/81.71
#33	BCE + 0.5Dice	21.32	29.02	99.17/84.76/91.75/90.53/93.01	99.47/89.67/94.55/93.66/95.47	91.75/70.65/82.80/**84.23**/81.42
#34	0.5BCE + Dice	21.32	29.02	99.19/85.10/91.95/91.25/92.67	99.57/91.49/95.56/94.31/96.84	91.53/69.14/81.75/80.46/83.09
#35	0.5BCE + 0.5Dice	21.32	29.02	99.18/84.83/91.79/90.54/93.08	99.53/90.79/95.17/94.08/96.29	91.99/70.37/82.61/80.67/**84.63**
#36	**BCE + Dice**	21.32	29.02	**99.19**/**85.16**/**91.98**/**91.65**/92.32	**99.57**/**91.62**/**95.63**/**94.41**/96.87	**92.13**/70.97/83.02/81.63/84.46

To further highlight the importance of the Laplacian pyramid operator in EGAM, we conduct a series of experiments comparing the impact of different edge operators on CD performance, including Isotropic Sobel, Krisch, Laplacian, and LoG operators in [79]. These operators extract features using predefined convolutional kernels during forward propagation. As their weights remain fixed and do not participate in backpropagation for gradient computation or weight updates, these operators do not increase model's Params or FLOPs. Specifically, the Isotropic Sobel detects edge changes by computing gradients in horizontal, vertical, and diagonal directions. In this study, eight distinct 3×3 Sobel kernels are utilized. The Kirsch sharing the same kernel size, exhibits a more uniform weight distribution across eight directions. The Laplacian highlights edges and texture changes by employing second-order derivatives. In this case, a 3×3 Laplacian kernel is applied, where the central pixel holds a negative weight, and the surrounding pixels are positive, enabling the weighted sum of pixel values to emphasize edges. Additionally, the LoG combines Gaussian smoothing with Laplacian operations, and a 5×5 LoG kernel is employed to suppress low-frequency information while enhancing edge features. The quantitative results of these operators are presented in Table IV (#18–#22), with visual comparisons shown in Fig. 18(f)–(j). It can be observed that these operators only partially retain edge features, with some information loss still present. Particularly when identifying small objects, the outermost edges exhibit varying degrees of false changes. In contrast, the EGAM utilizing the Laplacian pyramid operator demonstrates superior performance, with edge features closely aligning with the actual shape of the changes. Benefiting from its larger predefined convolution kernels, Gaussian smoothing for denoising, and adjacent scale disparity computation, EGAM effectively identifies blurred and weak edges, achieving optimal edge refinement and change enhancement. In contrast, the comparative operators with smaller kernels operating at a single scale generally exhibit weaker performance in CD.

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D. Discussion on the FGFM

The FGFM restores feature maps of different resolutions through the optical flow alignment module (FAM) and dynamically adjusts fusion weights using a GM, which automatically captures both global and local changes. Moreover, FGFM plays a crucial role in the alignment and fusion of feature maps at adjacent and cross-levels. Quantitative results in Table IV (#23–#25), show that the absence of any component leads to a performance degradation. Compared to adjacent fusion, the cross fusion is more critical. Small targets or local changes are often more pronounced in high-resolution feature maps, while large-scale environmental changes are more easily recognized in low-resolution feature maps. Adjacent fusion primarily relies on local information, thus limiting its ability to handle large or complex change areas. In contrast, cross fusion can comprehensively capture these changes. The visual results in Fig. 19(d)–(f) demonstrate an apparent change information loss when these key components are missing.

To further validate the optimality of our alignment strategy, we compare four mainstream alignment methods: bilinear interpolation (FAM1), unpooling (FAM2), deconvolution (FAM3), and two-channel flow alignment (FAM4). FAM1 aligns the feature maps of different resolutions using bilinear interpolation. FAM2 recovers the resolution via an unpooling operation, where the kernel size and stride are consistent with the ratio of high- to low-resolution feature channels in the FGFM input. FAM3 performs upsampling using a deconvolution layer, with the kernel size and stride set the same as in FAM2. Distinct from traditional methods, FAM4 directly generates two-channel optical flow for low-resolution features, representing vertical and horizontal displacement. Quantitative results are presented in Table IV (#26–#30), with corresponding visualizations in Fig. 19(g)–(k). On the LEVIR-CD and WHU-CD datasets, FAM3 and FAM4 demonstrated clear advantages over FAM1 and FAM2. While FAM1 performs well on the SYSU-CD dataset, its results remain inferior to our proposed FAM. Traditional alignment methods, such as FAM1, FAM2, and FAM3, each have certain limitations. Commonly used FAM1 relies solely on weighted averaging, which can lead to the loss of fine details, small targets, or edge information due to smoothing interpolation. FAM2 restores the maximum values from the pooling layer back to their original positions, but these maxima may not accurately represent the original features, leading to recovered feature maps that may contain unrealistic regions or loss of details. While FAM3 shows some improvement, deconvolution often introduces the "checkerboard effect," causing grid-like noise. In summary, these methods primarily rely on local information and often overlook spatial invariance. Unlike the four comparison methods that focus solely on unidirectional resolution recovery, our FAM generates four-channel optical flow for both low- and high-resolution features, enabling bidirectional synchronous alignment. It allows for more precise feature recovery, effectively preserving spatial details and reducing the loss of change information.

E. Discussion on the Coefficients of Loss Function

We employ a hybrid loss function combining BCE and Dice, and conduct experiments under various coefficient combinations, with the quantitative results shown in Table IV (#31–#36). The experimental results indicate that the BCE + Dice loss function achieved the best performance on both LEVIR-CD and WHU-CD datasets, outperforming the second-best 0.5BCE+ Dice loss function. On the SYSU-CD dataset, the highest IoU and F1 are obtained when using BCE alone, but the OA is relatively lower. This phenomenon is likely attributable to the BCE's limitations in addressing class imbalance and accounting for the influence of background regions, resulting in relatively poor classification for the background class. Despite the slightly lower IoU and F1 of BCE + Dice loss, it better fits the data distribution and demonstrates higher robustness. Therefore, the hybrid loss function with λ_1 and λ_2 both set to 1 exhibits strong generalization across all three datasets, providing a reliable strategy for CD.

VI. Conclusion

In this work, we propose an edge and flow guided iterative CNN for RS image CD. The network first utilizes an iterative VGG16 backbone with a Siamese structure to generate composite hierarchical features. This IB demonstrates superior feature extraction capabilities compared to a single backbone. After obtaining bi-temporal feature maps, the 3D-DEM is used to emphasize real discriminative changes while suppressing pseudo changes. The SimAM attention within this module interactively calculates the 3-D weights of neurons in the difference and concatenation branches, while maintaining a lightweight form. The EGAM, leveraging the Laplacian pyramid, further enhances high-frequency edge and refines change background, which is critical for CD in noisy and shadowed environments. Finally, the FGFM effectively aligns and propagates adjacent and cross-level fine-grained change features through optical flow estimation and warping, further optimizing the fusion process through the GM. Comparative experiments on three datasets validate the superiority of our method. Efficiency comparisons highlight the balance between Params and FLOPs. Ablation experiments and discussions underscore the contributions of each module and its key components to overall performance.

It is noteworthy that when our network adopts the iterative Mamba-T backbone, despite the increase in Params and FLOPs, its performance surpasses expectations. We believe that incorporating proven techniques from CNNs and transformers into Mamba architecture could further enhance its performance and broaden its applicability. Therefore, our future work will further explore the potential of Mamba-based CD methods. With the continuous progress of DL, exploring more lightweight and generalized networks for complex multiclass CD tasks will also be of significant importance.

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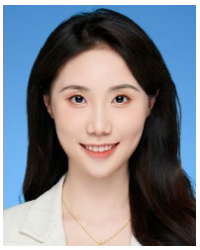
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